



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203: Software Engineering

System Documentation (SD)

Koperasi KADA Online System

Version 5

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Faculty of Computing

Prepared by:

Cheryl Cheong Kah Voon

Lau Yee Wen

Chau Ying Jia

Woo Cheng Shuan

Neo Li Xin

Revision Page

a. Overview

System Documentation serves as a document that describes all aspects of a system's operations, including system analysis, designing, development, implementation and testing. It serves as a centralized repository for information about the Koperasi KADA Online System including the system's requirements, design and testing processes. System Documentation provides a holistic view of these three primary components which are the SRS, SDD and STD. SRS outlines the functional and non-functional requirements of our system. SDD provides information about the system's architecture including the designs, structure and data flow. STD describes the testing methodology. This document ensures clarity and quality as well as creating better communication between all stakeholders, developers and operators throughout the development lifecycle.

b. Target Audience

The system documentation is designed for the project stakeholders such as the board of directors and admin users, who need to understand the system's functionalities and impact on operations. It is also intended for system developers and the quality assurance team to guide implementation and testing.

c. Project Team Members

Member Name	Role	Task	Status
Cheryl Cheong Kah Voon	Project manager	• Create user story for new user and member • Draw sequence diagram • Draw activity diagram	Partially complete
Lau Yee Wen	System analyst	• Document launch phase • Create user story for admin and board director • Draw sequence diagram • Draw activity diagram	Complete

Chau Ying Jia	System designer	- Design and plan the task - Go through the SD and get the important notes - Write the project overview and scopes	Complete
Neo Li Xin	System developer	- Draw state diagram - Draw class diagram - Draw a use case diagram	Partially complete
Woo Cheng Shuan	System testing	- List non-functional performance & other requirement - Write design constraints	Complete

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin	Partially completed all sections in the system documentation	30/11/2024
2.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin	- Complete session user story - Complete persona - Update the use case and domain model	1/12/2024
3.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin	- Updating team member roles - Reviewing user diagram - Updating ERD diagram, class model diagram, state diagram, activity diagram and sequence diagram - Reviewing document consistent phrasing and providing clearer system clarifications	28/12/2024
4.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin	- Reviewing user diagram - Updating ERD diagram, class model diagram, state diagram, activity diagram and sequence diagram	15/1/2025

		• Updating the test case	
5.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin	• Update Use Case • Updating ERD diagram, class model diagram, state diagram, activity diagram and sequence diagram • Updating the test case	12/2/2025

Note:

This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at School of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

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1. Introduction

The System Design Documentation (SDD) provides a detailed description of the system's architecture, components, and design solutions. It serves as a bridge between the requirements outlined in the System Requirements Specification (SRS) and the actual implementation of the system. This document outlines how the system will be structured, how its components will interact, and the technologies that will be used to ensure the system meets its intended purpose.

1.1 Purpose

The purpose of the System design documentation (SDD) is to provide a detailed architectural framework and design specifications for the Koperasi KADA System. It serves as a blueprint for developers which outlines the system's structural components, functional workflows and technical design elements required for implementation. The SDD bridges the gap between system requirements and implementation by translating the functional and nonfunctional requirements into a concrete design. It ensures clarity in design decisions, facilitates efficient development and supports alignment with project objectives. Moreover, the SDD aids stakeholders including administrators in understanding the system's design and how it addresses operational and technical needs.

1.2 Scope

The Koperasi KADA System is a web-based platform specifically designed to automate and optimize Koperasi KADA's key processes, focusing on member registration, loan applications and related administrative tasks. The system provides distinct functionalities for administrators and members, including member profile management, streamlines loan processing and comprehensive data reporting. The system enhances efficiency, accuracy and accessibility by reducing reliance on manual operations. It offers a unified solution to manage cooperative activities and decision-making effectively.

1. Member Registration

The employee can register for a membership by submitting the member application online to reduce the burden of manual data handling and processing.

2. Loan Application

The loan application can also be submitted online through the system.

3. Application status checking

This feature allows applicants to check the real-time status of their applications through the website. Once the application is approved, they will receive an automated email or SMS notification. The applications included membership applications and loan applications.

4. Check monthly and annual report

The system will generate monthly and annual reports for the Koperasi KADA Online System admin from the database, allowing the Koperasi KADA Online System admin to monitor their financial performance periodically. The system will also generate an annual report for the board directors to offer a comprehensive overview.

1.3 Definitions, Acronyms and Abbreviations

This section provides the definition of fall terms, acronyms and abbreviations used in this System Documentation.

- | | |
|---------|--|
| 1. KADA | Lembaga Kemajuan Pertanian Kemubu |
| 2. SD | System Documentation |
| 3. SRS | System Requirements Specification |
| 4. SDD | System Design Document |
| 5. STD | System Test Document |
| 6. CCTV | Closed-circuit Television |
| 7. ISO | International Organization for Standardization |
| 8. TLS | Transport Layer Security |
| 9. SSL | Secure Socket Layer |
| 10. CPU | Central Process Unit |
| 11. RAM | Random Access Memory |

1.4 References

- [1] Visual Paradigm. (2019). *What is Use Case Diagram?* Visual Paradigm.com.
<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-use-case-diagram/>

- [2] *What is Domain Class in UML?* (2024, February 29). GeeksforGeeks. <https://www.geeksforgeeks.org/what-is-domain-class-in-uml/>
- [3] Samir Bhimbha. (2024, April 18). *Understanding the Laravel MVC Architecture - Everything You Need to know 2024*. Pixlogix. <https://www.pixlogix.com/laravel-mvc-architecture/>

1.5 Overview

In this System Documentation, we will need testers to test that the system satisfies our design or technical requirements. We first introduce the five personas we are going to have throughout this section to let our development team have a clear understanding of the needs of users and to whom the system is designed to ensure that the system is well-developed. Next, we are going to explain our system features using a use case diagram, description, class diagram and state diagram. We are using sequence diagrams to help us to have a clearer view in this section. Lastly, we will finalize our research and make a conclusion on the aspect of system performances, requirements and design constraints.

2. Specific Requirements

In this section, we will identify the user characteristics, system features, use case details, system other requirements and system design constraints. These documents allow stakeholders, developers and operators to better understand the Koperasi KADA Online System main goal as well as the system's overall architecture.

2.1 Persona

The Koperasi KADA Online System software will be used by three main types of users: employees, members and admins. In this section, we will be using a persona to identify the characteristics of the users when meeting certain scenarios. These observations will allow us to make a better judgement about the system's functionality and design. Below are the personas we identified.

2.1.1 Persona 1 (User type 1/Actor 1)

User Need

A secretary managing membership applications.

User Stories

Sarah has been the secretary at KADA for over five years. Her main responsibility includes managing membership applications. Every week, she receives a stack of paper forms from potential members. Some are handwritten and barely legible, others are missing information, and a few arrive late or are misplaced altogether.

She spends hours manually sorting, verifying, and cross-checking details with incomplete spreadsheets. On a hectic day, Sarah needs to catch up on time, stuck between phone calls from applicants asking for updates and trying to find missing files in a growing pile on her desk. By the end of the day, her frustration peaks when she realises she accidentally approved a duplicate application and missed a deadline for another.

Sarah dreams of having a digital system where applicants can easily submit forms, validate their information, and track their application status. This would save her time, reduce errors, and allow her to focus on more strategic tasks. She wishes for a solution that's simple, efficient, and user-friendly—not just for her, but for the applicants as well.

2.1.2 Persona 2 (User type 2/Actor 2)

User Need

An archive staff ensures forms are stored securely and easily accessible.

User Stories

As a staff of KADA, Jonathan is tasked with managing and storing the physical forms in the organisation's archive. He has to make sure that the forms are stored in their respective places so that they are easily accessible by other staff.

One day Jonathan notices that part of the forms are missing. He tried asking other staff about the missing forms but to no avail, no one knew where they were. Confused and panicked, he checked the CCTV footage and confirmed that one of the staff did indeed steal the forms. Disappointed by the incident Jonathan reported it to the organization's management.

After acknowledging the potential security risks, Jonathan proposed to install a digital system where the forms can only be accessed online by authorized users. This system offers better management and security to the customer's information by introducing secure login features, thus reducing the risk of unauthorized users breaching their privacy.

2.1.3 Persona 3 (User type 3/Actor 3)

User Need

A customer service staff receiving necessary financial documents.

User Stories

Judie works in the customer service department. Every day at the counter she handles the physical forms submitted by the customers. Afterwards, the customers are required to submit their bank statements themselves as part of the loan application process. These bank statements are then sent to the management department for further validation.

As time went on, Judie noticed that some customers failed to bring the necessary documents to the next appointed date. This issue causes a delay in the validation process due to incomplete information provided. It also leads to customer dissatisfaction as their mistakes cost them a lot of time.

2.1.4 Persona 4 (User type 4/Actor 4)

User Need

An archive staff manager struggles to find space for paper storage.

User Stories

Kumar has worked as an archive staff manager for the past decade. He has been managing the physical reports and constantly checks on their organization, accuracy, and accessibility to ensure all documents are properly maintained and readily available when needed.

Back then, physical forms were much easier to handle with the help of other staff members. However, recently Kumar noticed that the increasing amount of paperloads each year are getting harder to manage. As applications increase, archive storage decreases as well as increasing workload for the staff members, causing fatigue and stress.

To address this issue, Kumar suggested implementing an online system for storing archives which would reduce manual labor, paper storage, and physical archiving while increasing productivity. This method also offers improved scalability in the future as the number of applications increases

2.1.5 Persona 5 (User type 5/Actor 5)

User Need

The board of directors manages the report.

User Stories

Ali is the board of directors of KADA Corporation Sdn Bhd. He is responsible for managing all types of reports, including member applications, loan applications and transaction records. The managing process involves viewing, checking, validating and printing. This time-consuming approach prompts Ali to seek a centralized and efficient solution for handling all reports.

One afternoon, during a particularly hectic day, Ali receives an urgent call from the CEO demanding a full account of loan approval. As he attempts to complete the report, he sees how much time is spent searching for papers, manually checking data, and ensuring everything is accurate. Frustrated, he thinks of a way to solve this issue.

A few days later, Ali thought of creating a dashboard that could access different types of reports in real-time. The system should display the reports, export them (PDF, Excel) and print a physical copy. Not to mention a feature that can automatically validate the reports to reduce human errors and also save time.

2.2 System Features

The objective of this Koperasi KADA Online System project is to design a system encompassing four key processes: member registration, loan application, reporting, and savings management. The system provides a means for users to do their membership and loan applications can be done online. It replaces the current manual processes, which are time-consuming, prone to errors, and heavily reliant on human resources. The system also ensures better management and security systems to optimise operations while ensuring data integrity and protection.

The system features are illustrated in Figure 2.2.1 below. The detailed description of each module and function is tabulated in Table 2.2.1.

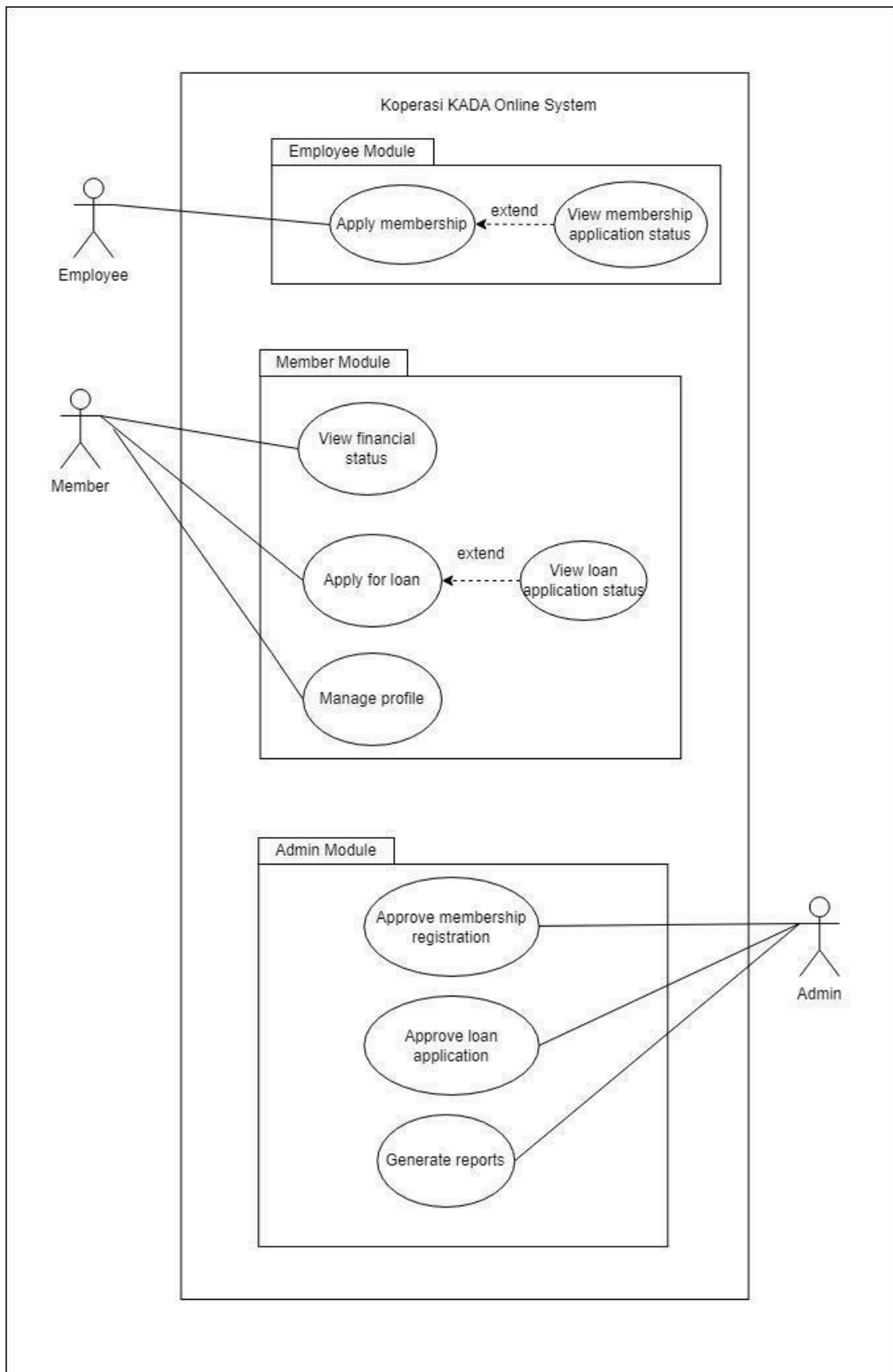


Figure 2.2.1: Use Case Diagram for KADA Online System

Table 2.2.1: Description of Module and Functions for Koperasi KADA Online System

Module	Function	Description
Employee Module	Apply membership	This use case describes Koperasi KADA employees applying for membership of Koperasi KADA.
Employee Module	View membership application status	This use case describes Koperasi KADA employees viewing their current membership application status, which is categorized into Approved, In Progress and Failed.
Member Module	View financial status	This use case describes Koperasi KADA members checking their financial account, recent transactions and loan earnings.
Member Module	Apply for loan	This use case allows Koperasi KADA members to register for the loan application.
Member Module	View loan application status	This use case describes Koperasi KADA members viewing their current loan application status, which is categorized into Approved, In Progress and Failed.
Member Module	Manage profile	This use case describes Koperasi KADA members managing their profile, including creating, updating and deleting their personal information.
Admin Module	Approve membership application	This use case describes Koperasi KADA board director and KADA Koperasi Online System admins to approve the membership application from the KADA Koperasi employees.
Admin Module	Approve loan application	This use case describes Koperasi KADA board director and KADA Koperasi Online System admins to approve the loan application from the KADA Koperasi members.
Admin Module	Generate reports	This use case describes Koperasi KADA board director and KADA Koperasi Online System admins to create reports of the successful member and loan applications from the Koperasi KADA employees and KADA Koperasi members respectively.

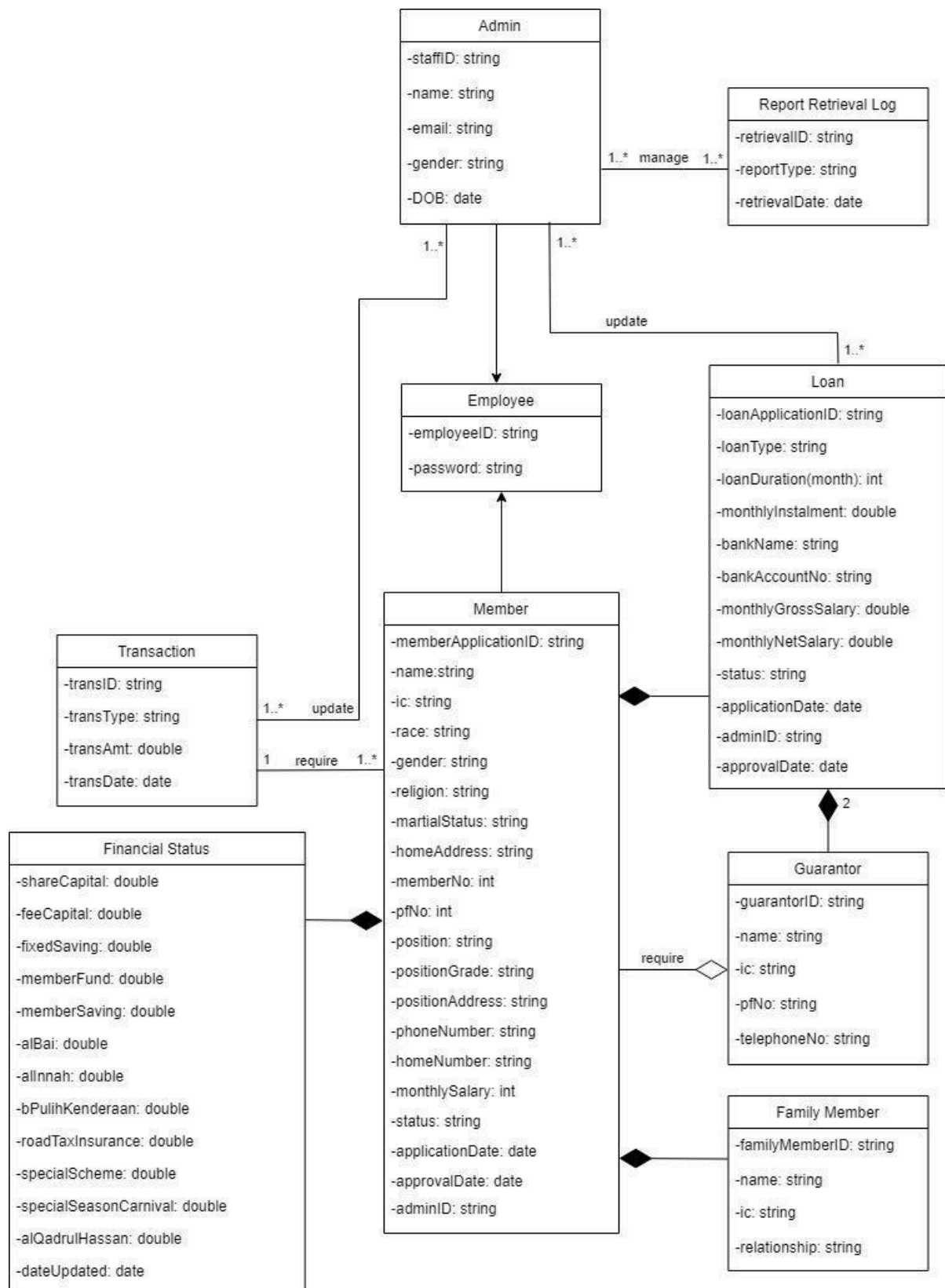


Figure 2.2.2: Domain Model for Koperasi KADA Online System

User Class

Represents all users that use the system. The information in this class will act as login credentials to the system.

- userID: A unique identifier for user
- password: Password to login system
- email: User email account

Member Class

Represents a person registered in the system as a member. As a user applies for a member, an object of the member is created.

- memberApplicationID: A unique identifier for membership application.
- name: Name of the member.
- ic: Identity card (MyKad) number of the member.
- gender: Gender of the member. This attribute is an enumeration with the values Male (Lelaki) and Female (Perempuan).
- religion: Religion of the member. This attribute is an enumeration with the values Islam, Buddhism (Buddha), Hinduism (Hindu), Christianity (Kristian) and Others (Lain-lain).
- race: Race of the members. This attribute is an enumeration with the values Malay (Melayu), Chinese (Cina), Indian (India) and Others (Lain-lain).
- maritalStatus: Marital status of the member.
- homeAddress: Address of the member. This includes line1, line2, Postcode (Poskod), City (Bandar) and State (Negeri).
- memberNo: Member number of the member. The member number would be available as the application is approved.
- pfNo: PF number of the member.
- position: Position held by the member.
- positionGrade: Position grade of the member.
- officeAddress: Address of the member on duty. This includes line1, line2, Postcode (Poskod), City (Bandar) and State (Negeri).
- phoneNumber: Phone number of the member.
- homeNumber: Home number of the member.
- monthlySalary: Monthly salary of the member with the currency of RM.
- status: Application status of the member. This attribute is an enumeration with the values Processing (Sedang Diproses), Rejected (Ditolak) and Accepted (Diterima).
- applicationDate: Date when the application is submitted.

The Member class is inherited from the User class. A member whose status is accepted would be eligible to apply for a loan.

Financial Status Class

Represents the financial status of the associated member.

- **shareCapital:** Share capital (Modah Syer) of the associated member with the currency of RM.
- **feeCapital:** Fee Capital (Modal Yuran) of the associated member with the currency of RM.
- **fixedSaving:** Fixes Saving or Fixed Deposit (Simpanan Tetap) of the associated member with the currency of RM.
- **memberFund:** Member Fund (Tabung Anggota) of the associated member with the currency of RM.
- **memberSaving:** Member Saving (Simpanan Anggota) of the associated member with the currency of RM.
- **alBai:** The loan amount in the Al-Bai category for the associated member in RM.
- **alInnah:** The loan amount in the Al-Innah category for the associated member in RM.
- **bPulihKenderaan:** The loan amount in the Baik Pulih Kenderaan category for the associated member in RM.
- **roadTaxInsurance:** The loan amount in the Road Tax & Insurance category for the associated member in RM.
- **specialScheme:** The loan amount in the Special Scheme (Khas) category for the associated member in RM.
- **specialSeasonCarnival:** The loan amount in the Special Season Carnival (Karnival Musim Istimewa) category for the associated member in RM.
- **alQadrulHassan:** The loan amount in the Al-Qadrul Hassan category for the associated member in RM.
- **dateUpdated:** The last date of the financial status is updated. The update is made by the admin and detailed in the Transaction class.

The Financial Status class is a composition of the Member class. The relationship is a one to one relationship.

Loan Class

Represents the loan application applied by the associated member.

- **loanApplicationID:** A unique identifier for the loan application.
- **loanType:** The type of loan applied by the member. This attribute is an enumeration with the values Al-Bai, Al-Innah, Baik Pulih Kenderaan, Road Tax and Insurance, Special Scheme (Khas), Special Season Carnival (Karnival Musim Istimewa) and Al-Qadrul Hassan.
- **appliedLoan:** The amount of loan applied by the member in RM.
- **loanPeriod:** The instalment period of the applied loan.
- **monthlyInstalment:** The monthly instalment amount of the applied loan in RM.
- **bankAccountNo:** The bank account number of the applicant.
- **bankName:** The bank associated with the bank account number provided.
- **monthlyGrossSalary:** The monthly gross salary of the applicant in RM.
- **monthlyNetSalary:** The monthly net salary of the applicant in RM.
- **signature:** The reference to the uploaded acknowledgement form with the signature of the applicant's employer.

- status: Application status of the member. This attribute is an enumeration with the values Processing (Sedang Diproses), Rejected (Ditolak) and Accepted (Diterima).
- applicationDate: Date when the application is submitted.

A member of the cooperative fulfilling certain criterias would be able to apply for loans.

(tbc..)

Figure 2.2.3 shows the state diagram for the member application process. After submission, it will be checked and validated by the admin. When successful, a report will be generated by the board of directors and the status of the new user will be updated.

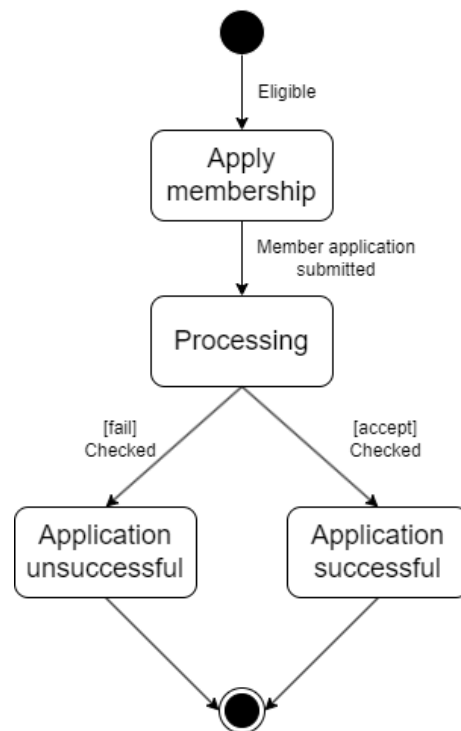


Figure 2.2.3: State Diagram for Member Application Koperasi KADA Online System

Figure 2.2.4 shows the state diagram for the loan application process. During this process, the person applying must be a KADA member first and foremost. After the loan application submission, it will be checked and validated by the admin. When successful, a report will be generated by the board of directors and the status of members will be updated.

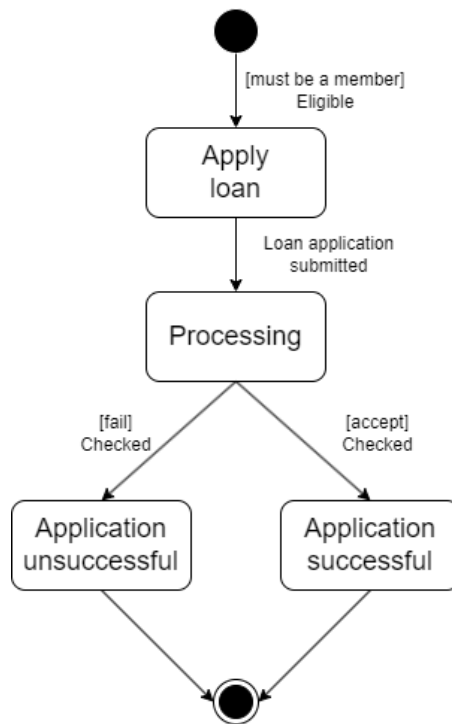


Figure 2.2.4: State Diagram for Loan Application Koperasi KADA Online System

2.3 Launch Phase

The project's components and modules are listed in the product backlog according to priority. Table 1.2 shows the product backlog for the Kada System.

Sprint	Use Case Involved	Team Members Assigned
<i>Sprint #1</i>	UC001 Register Member	Lau Yee Wen
<i>User Module</i>	UC002 View Membership Application Status	
<i>Sprint #2</i>	UC003 View financial status	Cheryl Cheong Kah Voon Neo Li Xin
<i>Member Module</i>	UC004 Apply for a loan	
	UC005 View Loan Application Status	
	UC006 Manage Profile	
	UC007 Terminate Membership Application	

<i>Sprint #3</i>	UC008 Approve Membership Application	Chau Ying Jia
<i>Admin Module</i>	UC009 Approve Loan Application	Woo Cheng Shuan
	UC0010 Approve Terminate Membership Application	
	UC0011 Generate Reports	

Use Case Details

2.4.1 UC001: Use Case < Apply Membership >

Table 2.4.1.1: User Story Description for Apply Membership

User story: Apply Membership
ID: US001
User Story Description: As a Koperasi KADA employee I want to register as a member of the cooperative So that I can enjoy more benefits that are provided by the cooperative.
Flow of events: 1. The employee accesses the system by searching the website. 2. The employee chooses the “Permohonan Anggota ” on the navigation bar. 3. The employee fills in all the personal details.

4. The employee clicks the submit button to submit the application form. 5. The system will prompt a message to confirm the submission.
Alternative flow: 1. If the personal information is incomplete, the system will prompt a message to remind the employee to fill in the incomplete place. 2. If the application is rejected, the employee must reapply and cannot log in to the system.
Acceptance Criteria Precondition: 1. The employee must be a current employee of the Koperasi KADA. Postcondition: 1. If the application is approved, the member will receive a notification and the employee can log into the system to enjoy the benefits provided by the cooperative. 2. If the application is rejected, the employee will receive a notification. The employee must reapply and cannot log in to the system. Other conditions: 1. The new member details will be stored in the member database. 2. A rejected application must clearly notify the reason for the rejection so that the employee can easily reapply.
Exception flow: 1. If the system is down, the system will prompt a message to advise the employee to try later. 2. If the employee has invalid details, the system will prompt a message to remind the employee to re-enter.

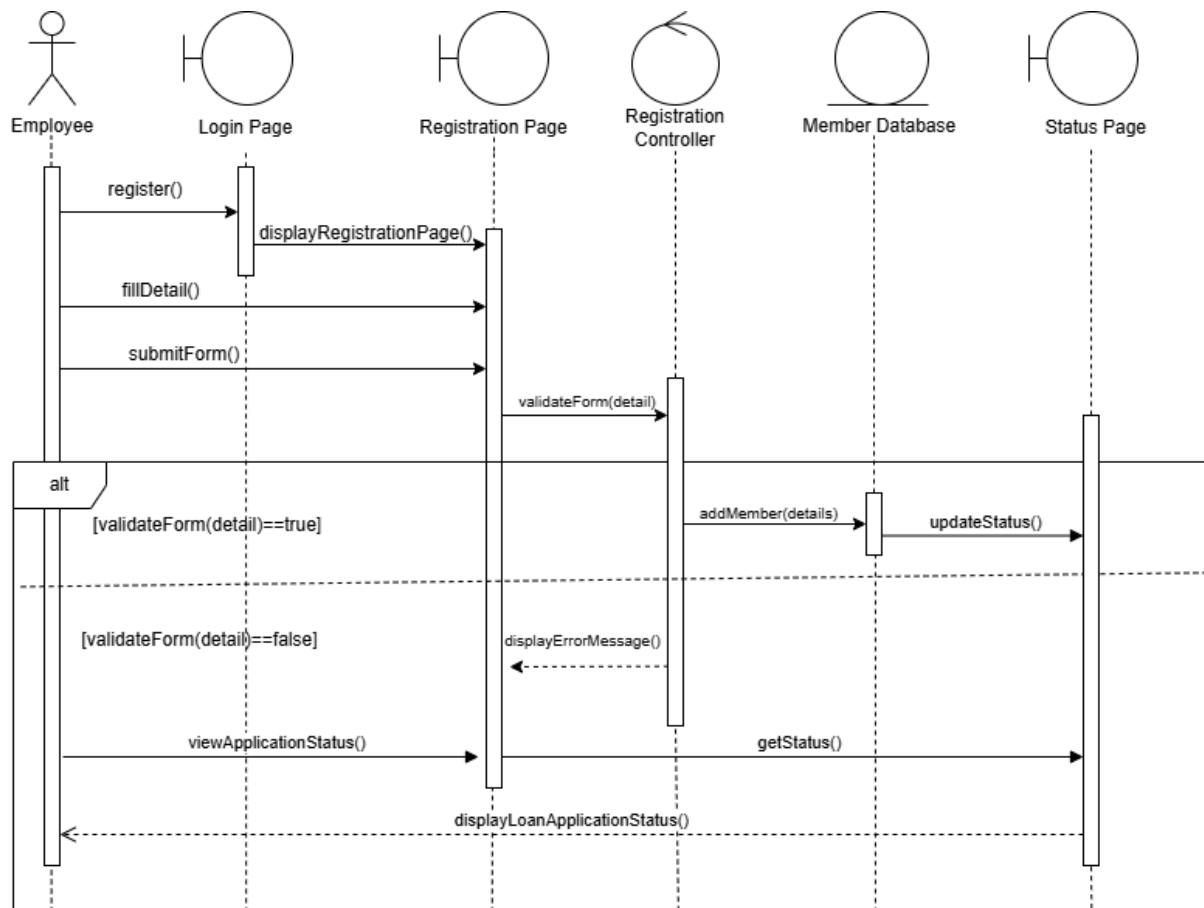


Figure 2.4.1.2: Sequence Diagram for Apply Membership

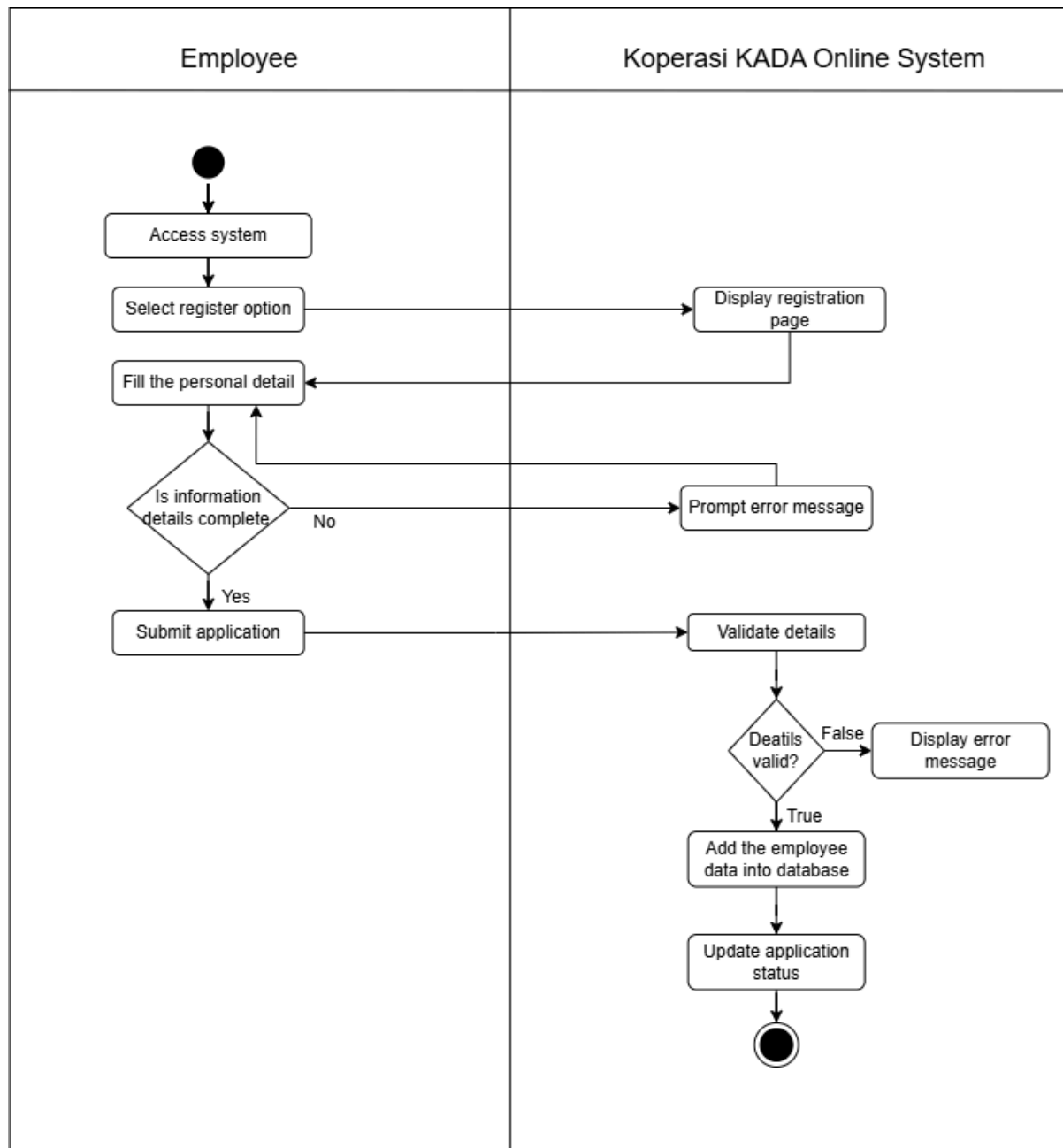


Figure 2.4.1.3: Activity Diagram for Apply Membership

2.4.2 UC002: Use Case < View Membership Application Status >

Table 2.4.2.1: User Story Description for View Membership Application Status

User story: View Membership Application Status
ID: US002
User Story Description: As a Koperasi KADA employee I want to view my current membership application status So that I can keep track of my latest membership application status.
Flow of events: 1. The employee accesses the system by searching the website. 2. The employee selects “Perkhidmatan” on the top panel, then selects “Permohonan Anggota” and lastly selects “Status Permohonan Anggota”. 3. The system dashboard will display the current status, ranging from step 1 to 4: • Step 1: Form submitted • Step 2: Application is currently under process • Step 3: Application succeed/fail • Step 4: Application result notification.
Alternative flow: -
Acceptance Criteria Precondition: 1. The employee must already submit the membership application. Postcondition: 1. If the application is approved, the member will receive a notification and the employee can log into the system. 2. If the application is rejected, the employee will receive a notification. The employee must reapply and cannot log in to the system. Other conditions: 1. The new member details will be stored in the member database. 2. A rejected application must clearly notify the reason for the rejection so that the employee can easily reapply.

Exception flow:

1. If the system is down, the system will prompt a message to advise the employee to try later.
2. If the employee has invalid details, the system will prompt a message to remind the employee to re-enter.

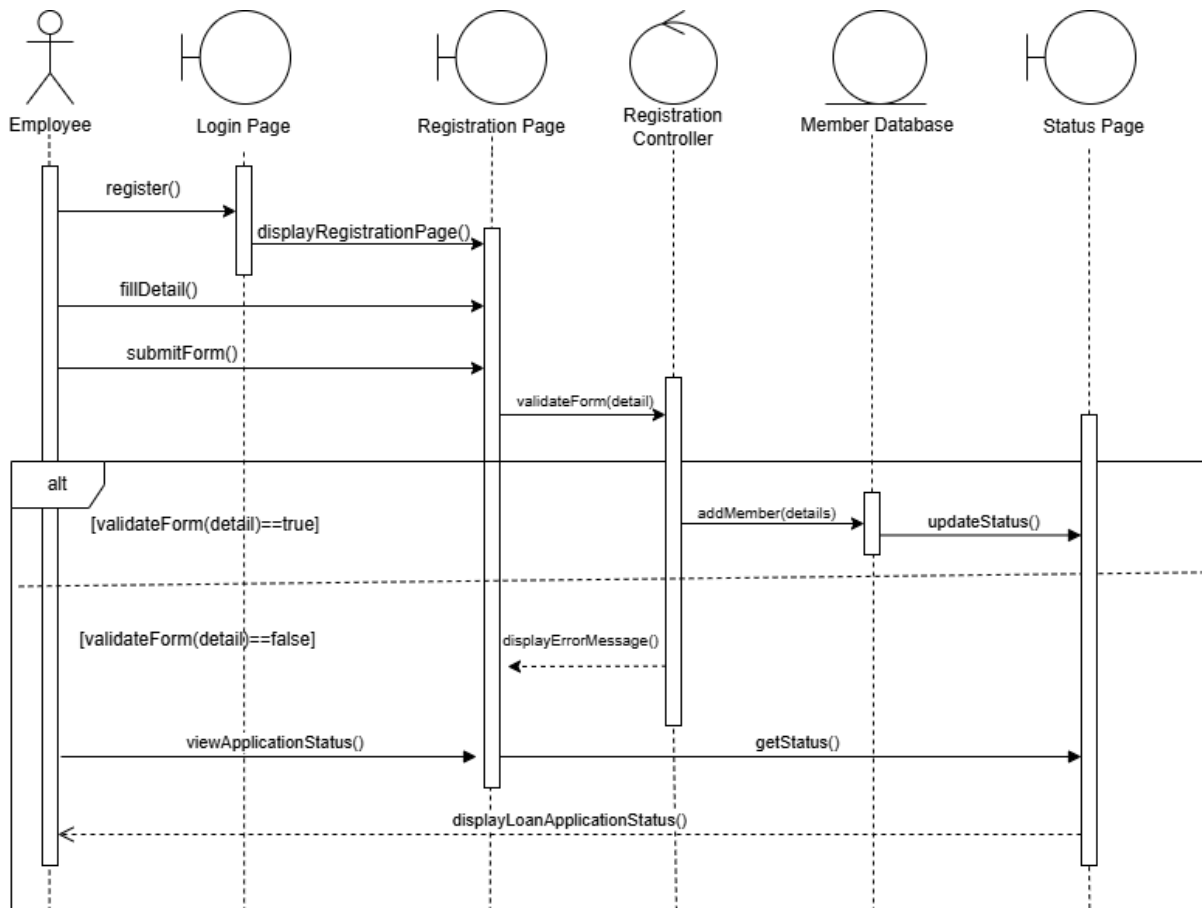


Figure 2.4.2.2: Sequence Diagram for View Membership Application Status

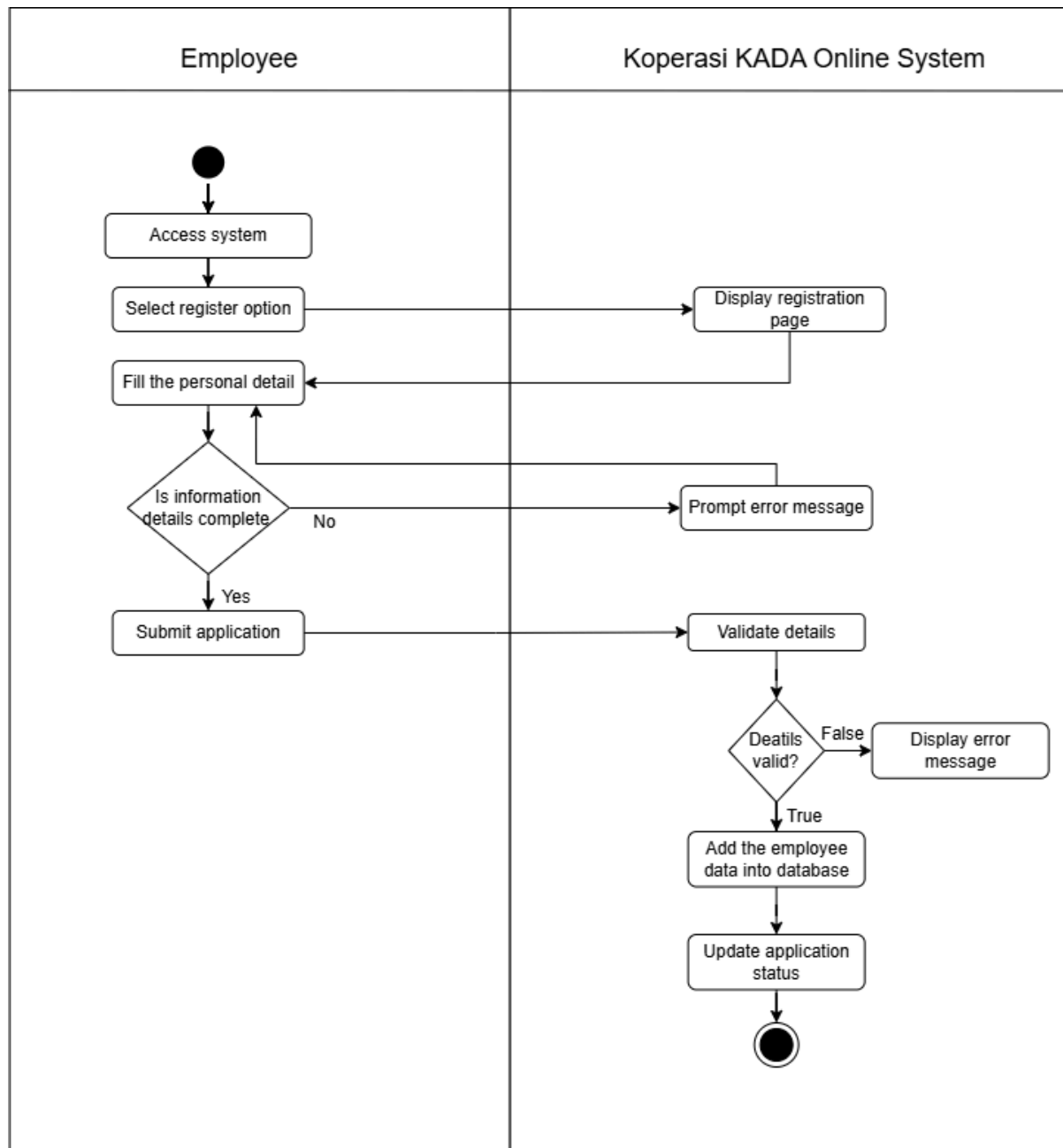


Figure 2.4.2.3: Activity Diagram for View Membership Application Status

2.4.3 UC003: Use Case < View Financial Status >

Table 2.4.3.1: User Story Description for View Financial Status

User story: View Financial Status
ID: US003
User Story Description:

As a Koperasi KADA member I want to view my financial status So that I can get to know my current financial status.
Flow of events: 1. The member logs in to the system by using the registered email. 2. The member navigates to the “Profile” section. 3. The member reviews the financial status.
Alternative flow: -
Acceptance Criteria Precondition: 1. The member logs in to the system. Postcondition: 1. The member successfully views the financial information. Other conditions: -
Exception flow: 1. If the system is down, the system will prompt a message to advise the employee to try later.

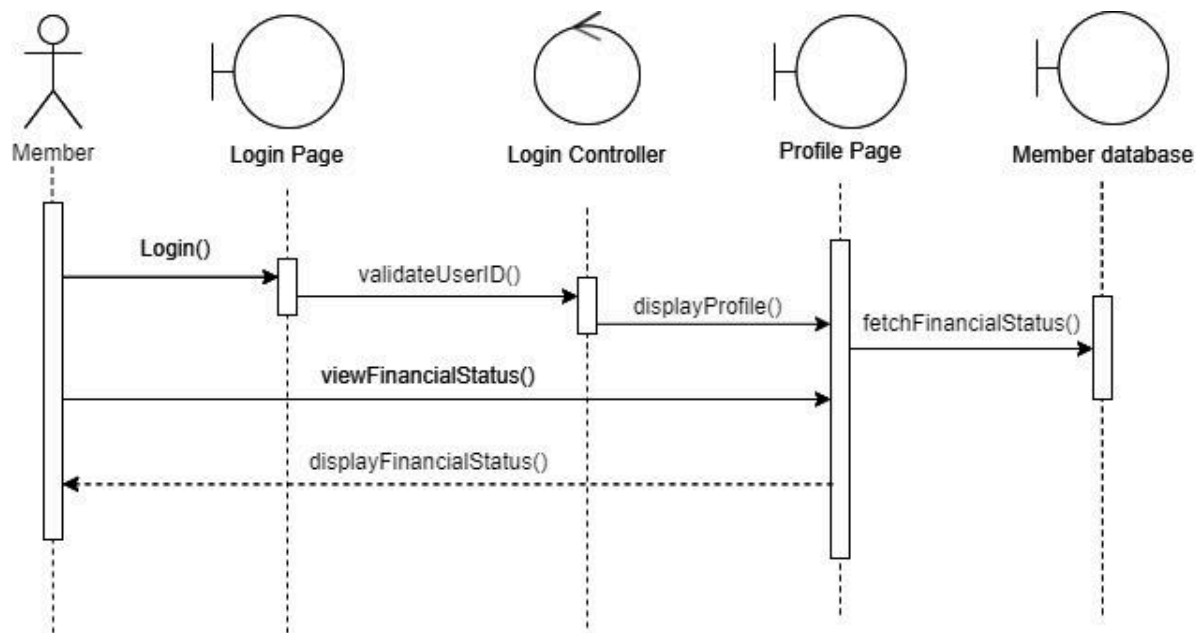


Figure 2.4.3.2: Sequence Diagram for View Financial Status

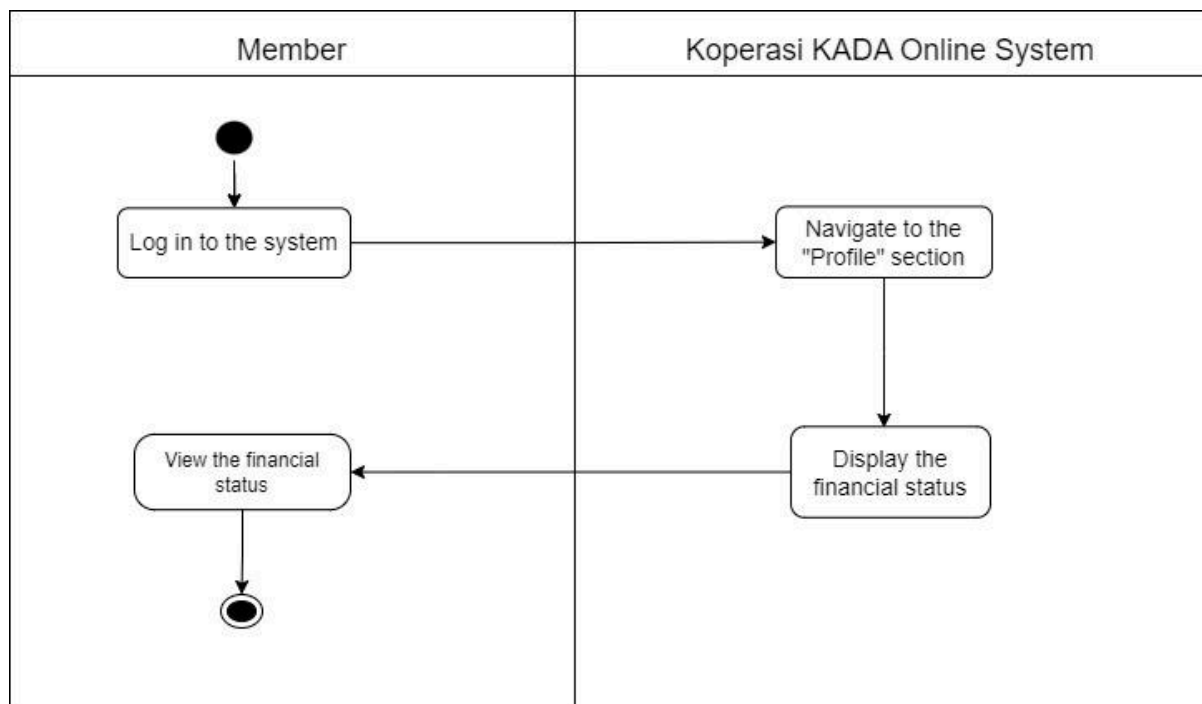


Figure 2.4.3.3: Activity Diagram for View Financial Status

2.4.4 UC004: Use Case < Apply for Loan >

Table 2.4.4.1: User Story Description for Apply for Loan

User story: Apply for Loan
ID: US004
User Story Description: As a Koperasi KADA member I want to apply for a loan from the cooperative So that I can get the
Flow of events: 1. The member logs in to the system by using the registered email. 2. The member chooses the “Permohonan Pembiayaan Anggota” on the navigation bar. 3. The member fills in all the personal details in the form. 4. The member needs to upload the required documents such as "Butir-Butir Penjamin" and "Borang Pengesahan Majikan". 5. The member presses the submit button to submit the application form. 6. The system will prompt a message to confirm the submission.
Alternative flow: 1. If the personal information is incomplete, the system will prompt a message to remind the employee to fill in the incomplete place. 2. If the application is rejected, the employee must reapply and cannot log in to the system.
Acceptance Criteria Precondition: 1. The member logs in to the system. Postcondition: 1. If the application is approved, the member will receive a notification and the employee can log into the system to enjoy the benefits provided by the cooperative. 2. If the application is rejected, the employee will receive a notification. The employee must reapply and cannot log in to the system. Other conditions:

1. A rejected application must clearly notify the reason for the rejection so that the employee can easily reapply.

Exception flow:

1. If the system is down, the system will prompt a message to advise the employee to try later.
2. If the employee has invalid details, the system will prompt a message to remind the employee to re-enter.

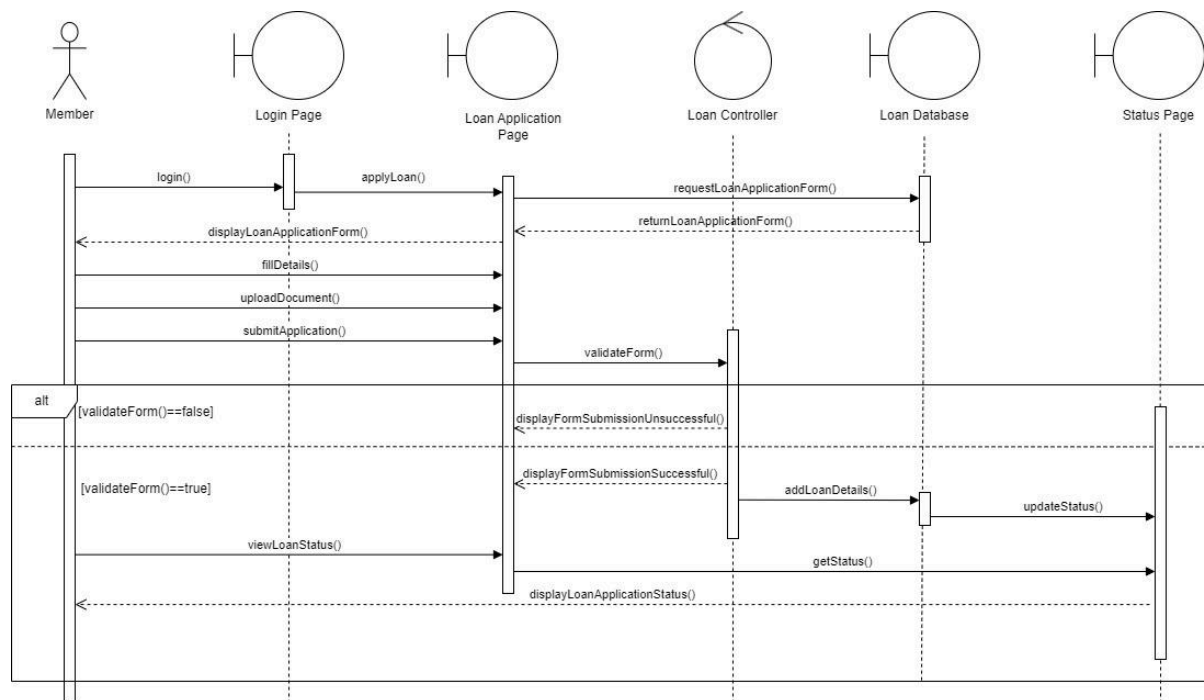


Figure 2.4.4.2: Sequence Diagram for Apply for Loan

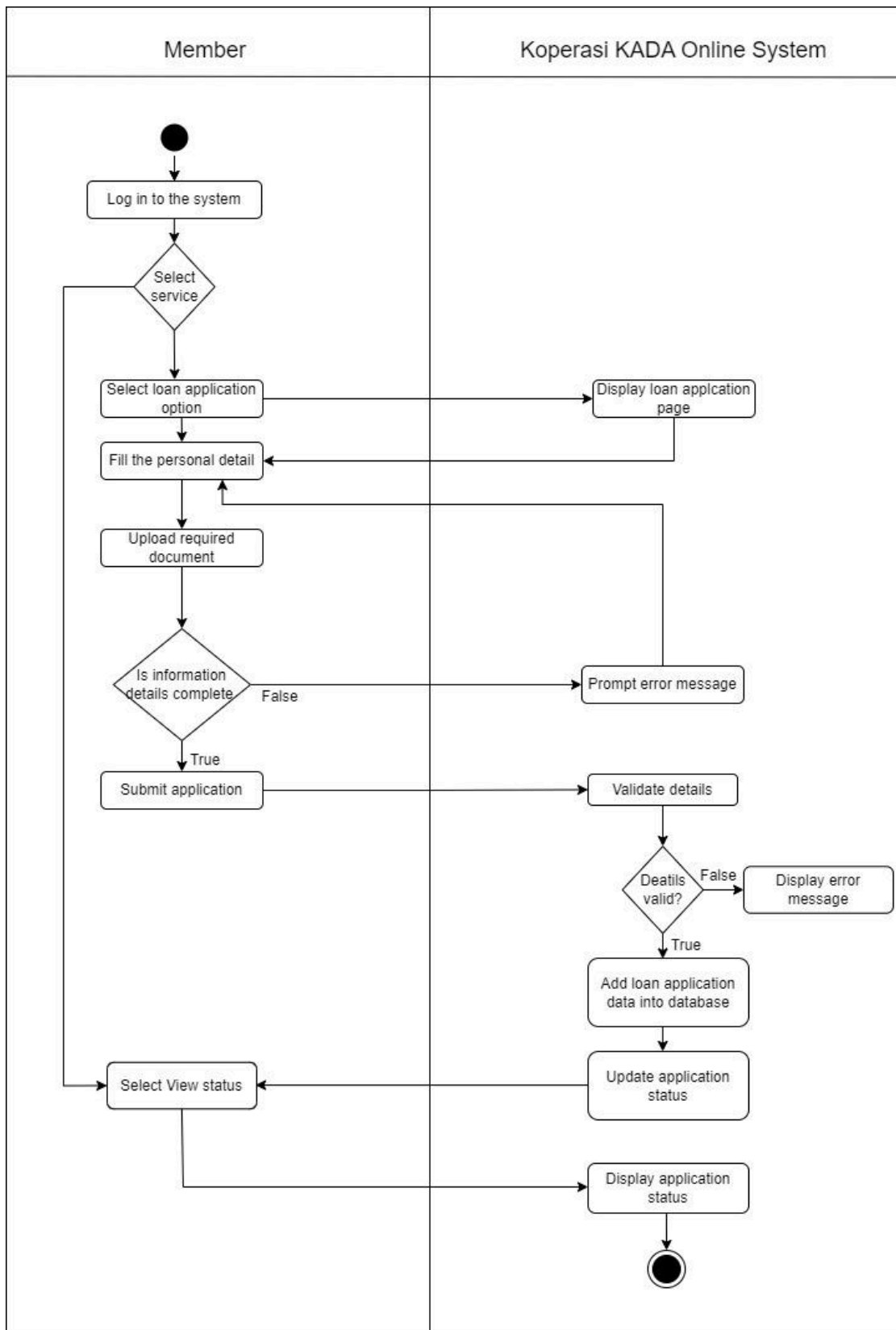


Figure 2.4.4.3: Activity Diagram for Apply for Loan

2.4.5 UC005: Use Case < View Loan Application Status >

Table 2.4.5.1: User Story Description for View Loan Application Status

User story: View Loan Application Status
ID: US005
User Story Description: As a Koperasi KADA member I want to view my loan application So that I can monitor the progress and take necessary action if needed.
Flow of events: 1. The employee accesses the system by searching the website. 2. The member logs in to the system by using their registered email. 3. The employee selects the profile page 4. The employee clicks the submit button to submit the application form. 5. The system will prompt a message to confirm the submission.
Alternative flow: 1. If the personal information is incomplete, the system will prompt a message to remind the employee to fill in the incomplete place. 2. If the application is rejected, the employee must reapply and cannot log in to the system.
Acceptance Criteria Precondition: 1. The employee must be an active employee of the Koperasi KADA. Postcondition: 1. If the application is approved, the member will receive a notification and the employee can log into the system to enjoy the benefits provided by the cooperative. 2. If the application is rejected, the employee will receive a notification. The employee must reapply and cannot log in to the system. Other conditions: 1. The new member details will be stored in the member database.

2. A rejected application must clearly notify the reason for the rejection so that the employee can easily reapply.

Exception flow:

3. If the system is down, the system will prompt a message to advise the employee to try later.
4. If the employee has invalid details, the system will prompt a message to remind the employee to re-enter.

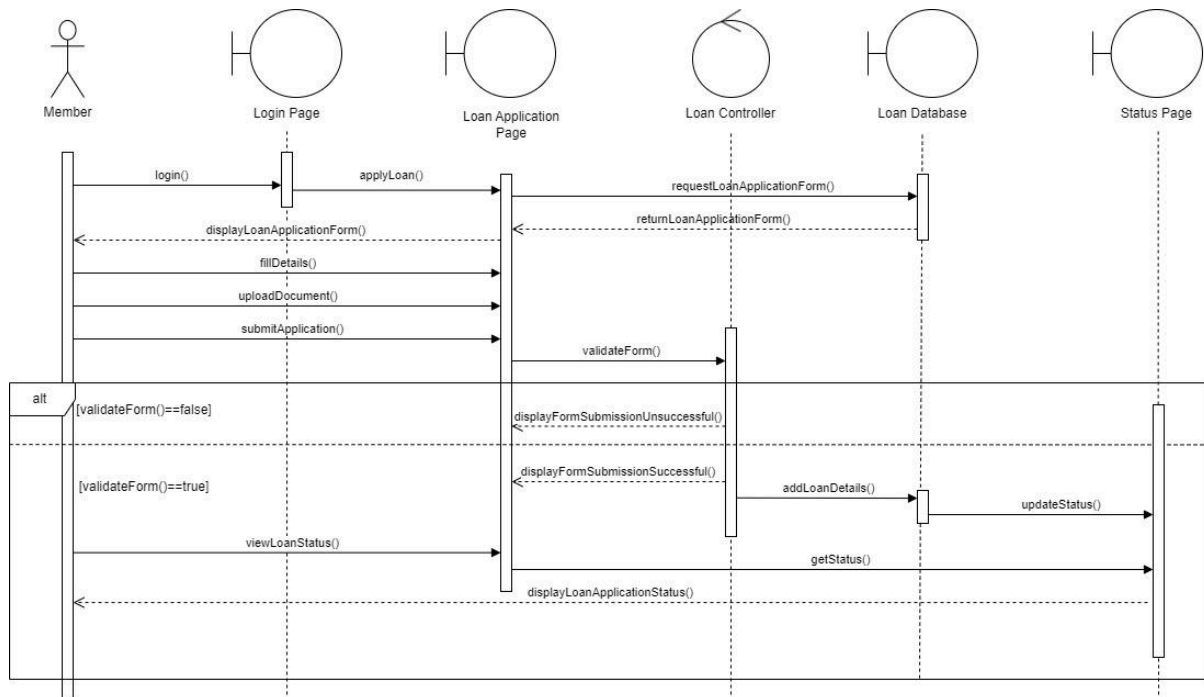


Figure 2.4.5.2: Sequence Diagram for View loan application status

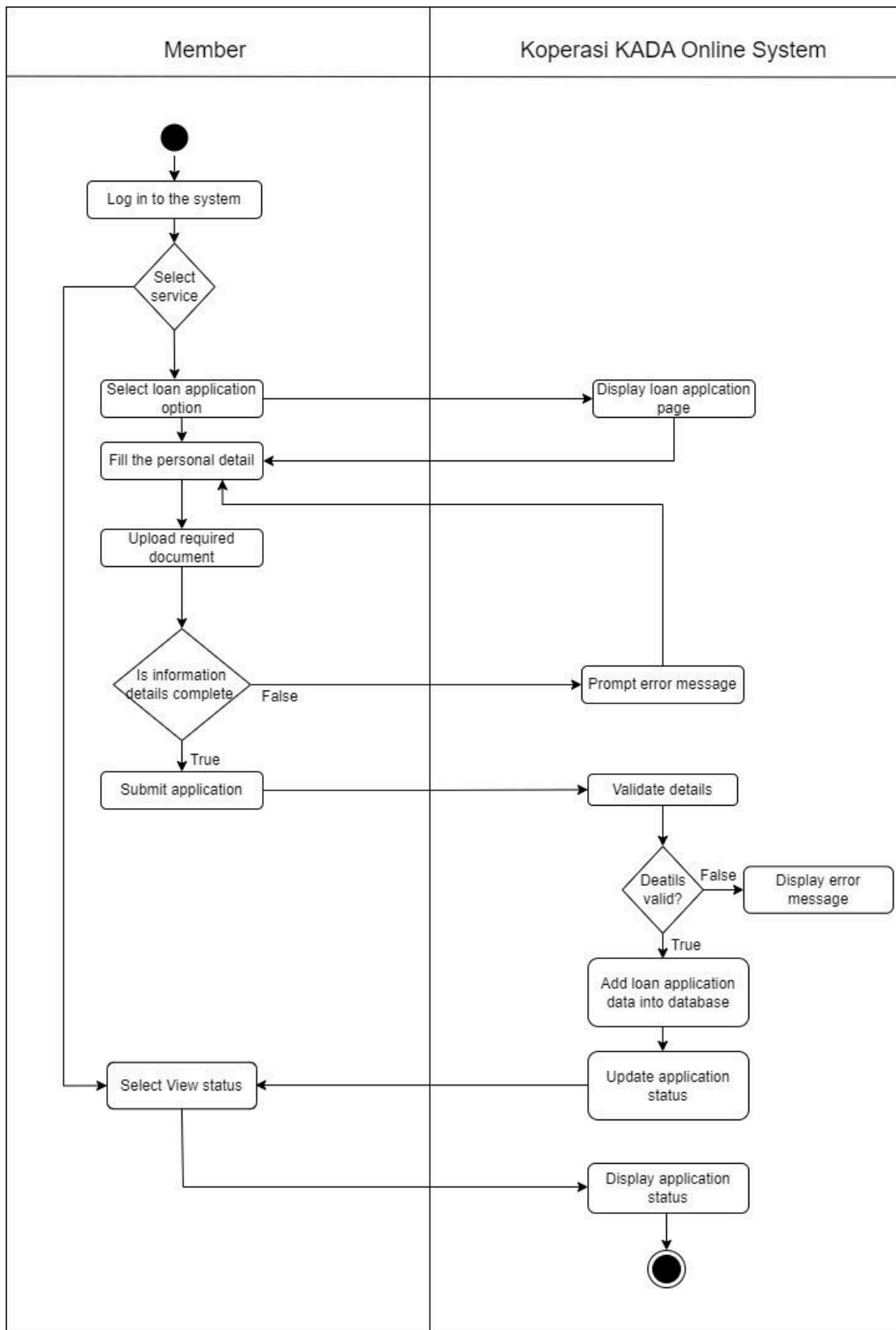


Figure 2.4.5.3: Activity Diagram for View loan application status

2.4.6 UC006: Use Case < Manage Profile >

Table 2.4.6.1: User Story Description for Manage Profile

User story: Manage Profile
ID: US006
User Story Description As a Koperasi KADA member I want to manage my profile So that I can ensure my personal details and activities with the cooperative are accurate and up to date.
Flow of events: 1. The member logs into the system using their registered email 2. The member navigates to the profile section. 3. The profile will display the member's profile details include 3.1. personal information 3.2. loan application details and status 3.3. contribution history 4. The member can update any desired profile information and click the save button. 5. The system will validate the updated information. 6. If all input is valid: 6.1. The system will update the information in the database 6.2. The system will prompt a success message. 7. If the input is invalid, the system will prompt an error message to inform the user to update the information again.
Alternative flow <i>n</i>: 1. If the member enters invalid data, the system will request the member to reenter the data. 2. If the system is temporarily unavailable, it will display a maintenance message and ask the member to try again.
Acceptance Criteria Precondition: 1. The member logged in to the system. Postcondition: 1. Profile updates are successfully saved in the database. 2. Members can view the updated information in their profile immediately. Other Conditions: 1. The system must validate all inputs before saving.
Exception flow:

1. If the system is down, the system will prompt a message to advise the employee to try later.

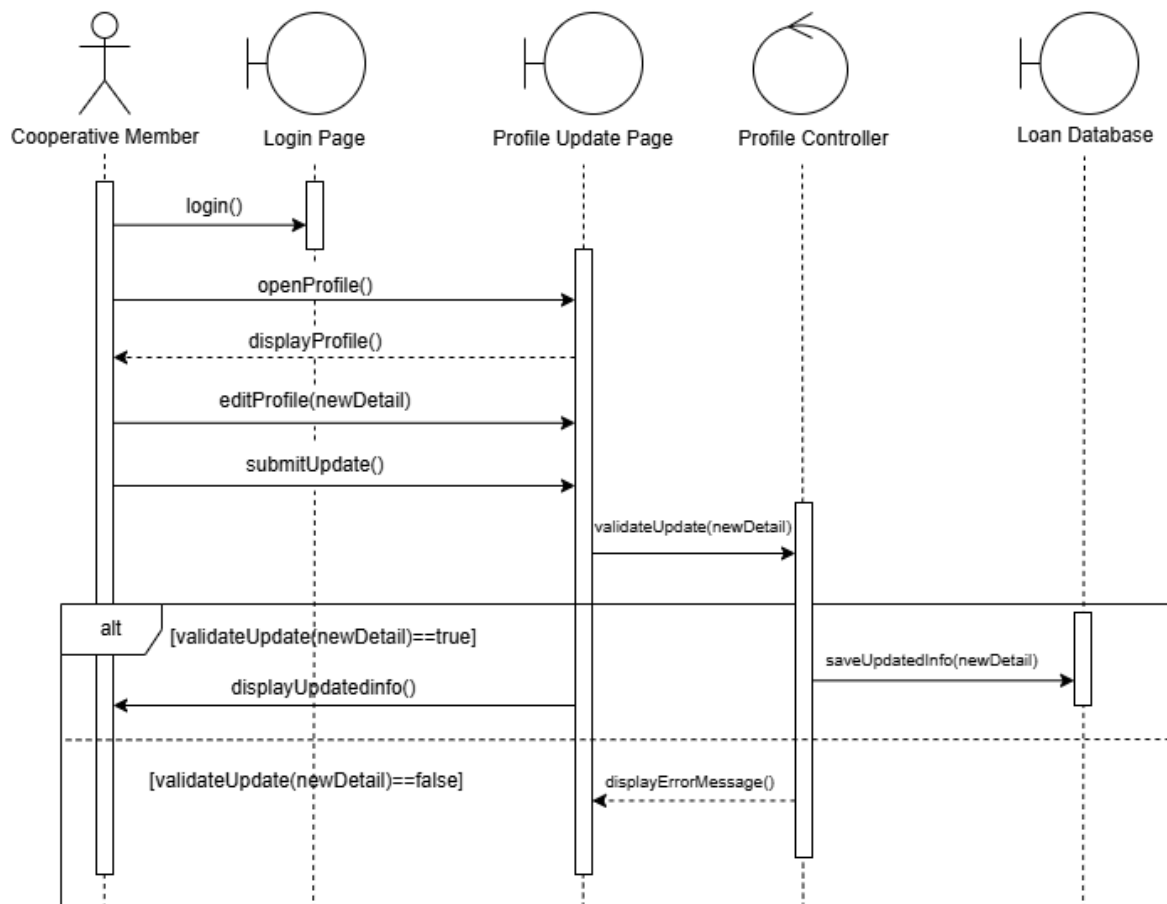


Figure 2.4.6.2: Sequence Diagram for Manage Profile

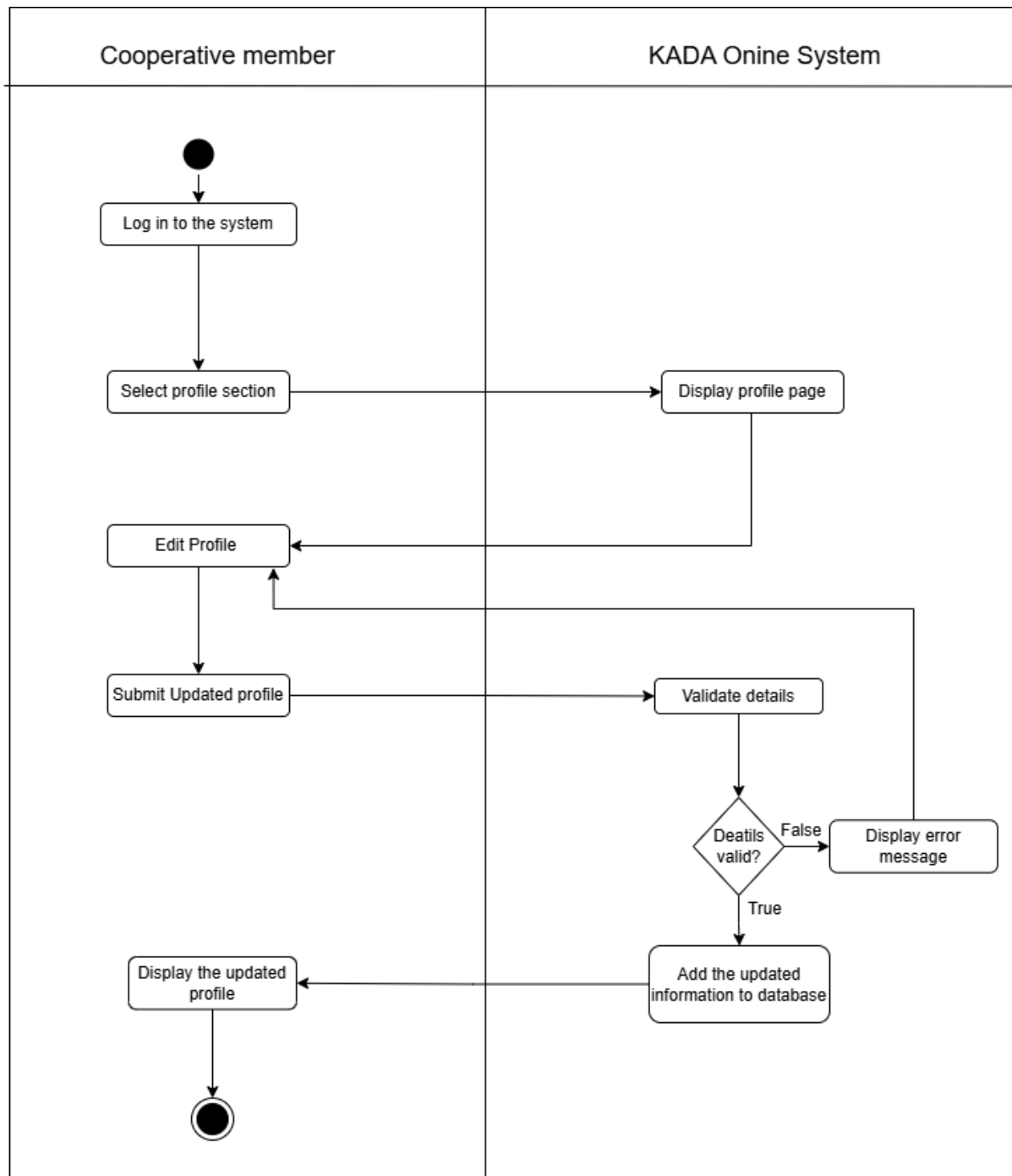


Figure 2.4.6.3: Activity Diagram for Manage Profile

2.4.7 UC007: Use Case < Terminate the Membership>

Table 2.4.7.1: User Story Description for Terminate the Membership

User story: Terminate the Membership
ID: US007
User Story Description As an Koperasi KADA member I want to terminate the membership So that I can discontinue my association with Koperasi KADA and avoid any further obligations or responsibilities.
Flow of events: 1. The member logs into the system. 2. The member navigates to the "Permohonan berhenti menjadi anggota" section. 3. The system displays the termination form 4. The member fills in the required details and submits the request. 5. The system validates the request and sends it to the admin for approval. 6. The admin reviews and approves/rejects the termination request. 7. The system update the status of termination.
Alternative flow <i>n</i>: -
Acceptance Criteria Precondition: 1. The member must have an active membership. Postcondition: 1. The termination request is submitted, and the member is notified of the approval or rejection status.

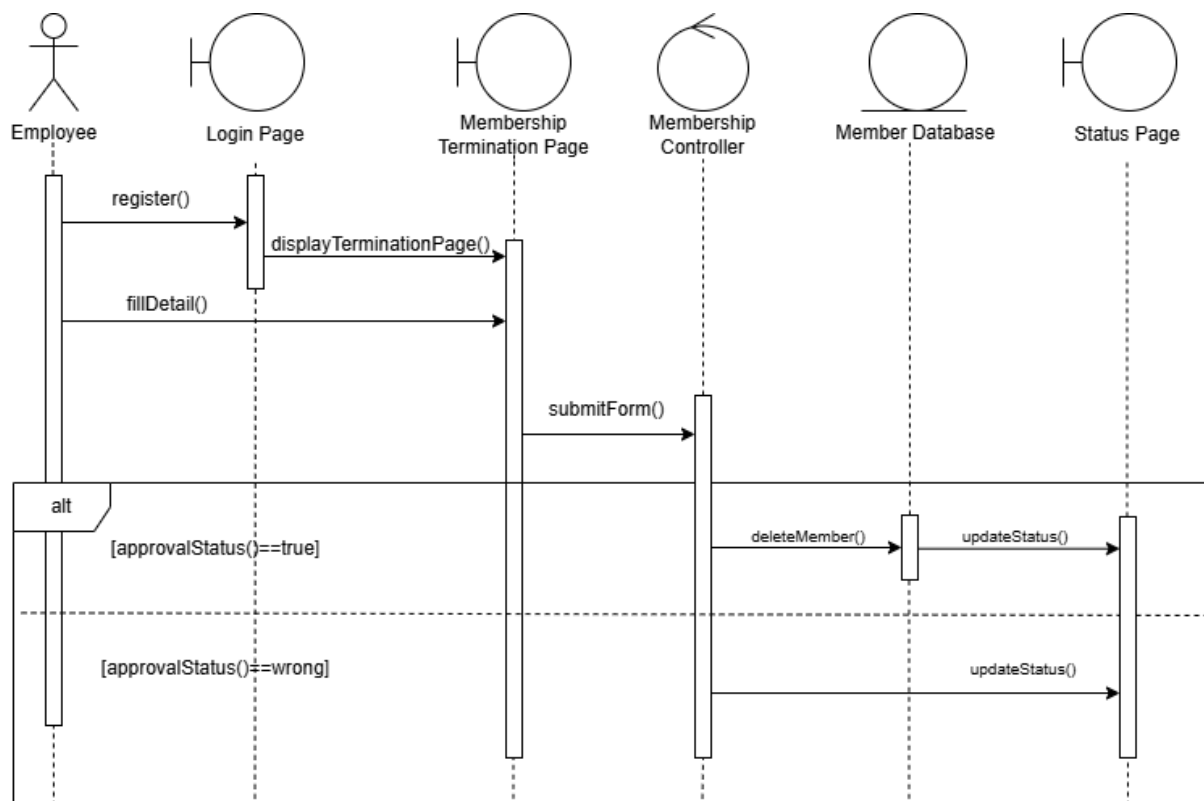


Figure 2.4.7.2: Sequence Diagram for Terminate the Membership

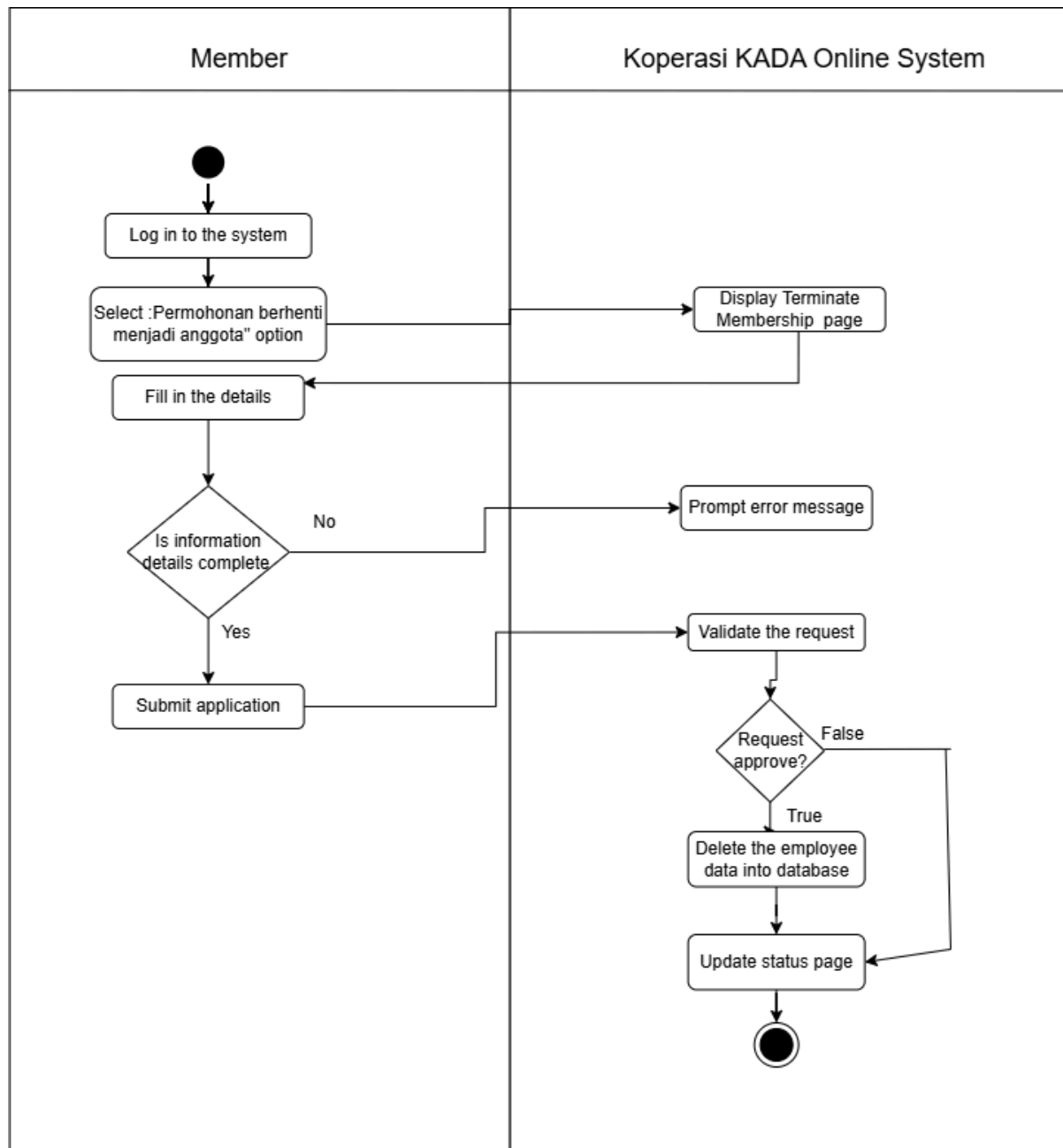


Figure 2.4.7.3: Activity Diagram for Terminate the Membership

2.4.8 UC008: Use Case < Approve Membership Application >

Table 2.4.8.1: User Story Description for Approve Membership Application

User story: Approve Membership Application
ID: US008
User Story Description As an Koperasi KADA admin I want to view and validate the membership application information So that I can approve the Koperasi KADA employees' membership application.
Flow of events: - The admin logs into the system. - The admin selects the three status of the membership application progress, either approved, in progress and fail. - The system retrieves relevant data from the database, including new, in progress and completed member applications. - The system generates an overall member application report summary and detailed breakdown of each member's information in a specified period. - The system displays the report, providing options to: view the report in detail, export the report (PDF, Excel) or print the report.
Alternative flow <i>n</i>: If the applicant didn't provide the accurate information, the admin will ask the applicant to update new information.
Acceptance Criteria Precondition: - The admin logged in to the system. Postcondition: - Accurate and timely reports are generated. - Visual charts and summaries are easy to understand. - Exportable to formats like PDF or Excel.

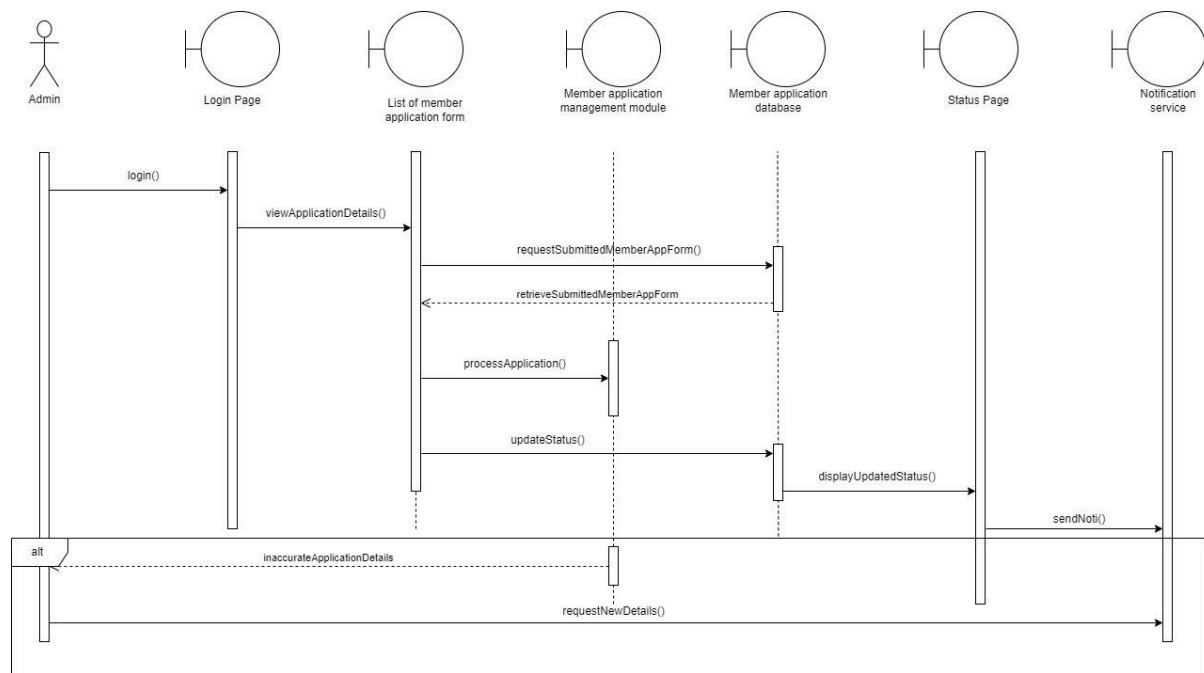


Figure 2.4.8.2: Sequence Diagram for Approve Membership Application

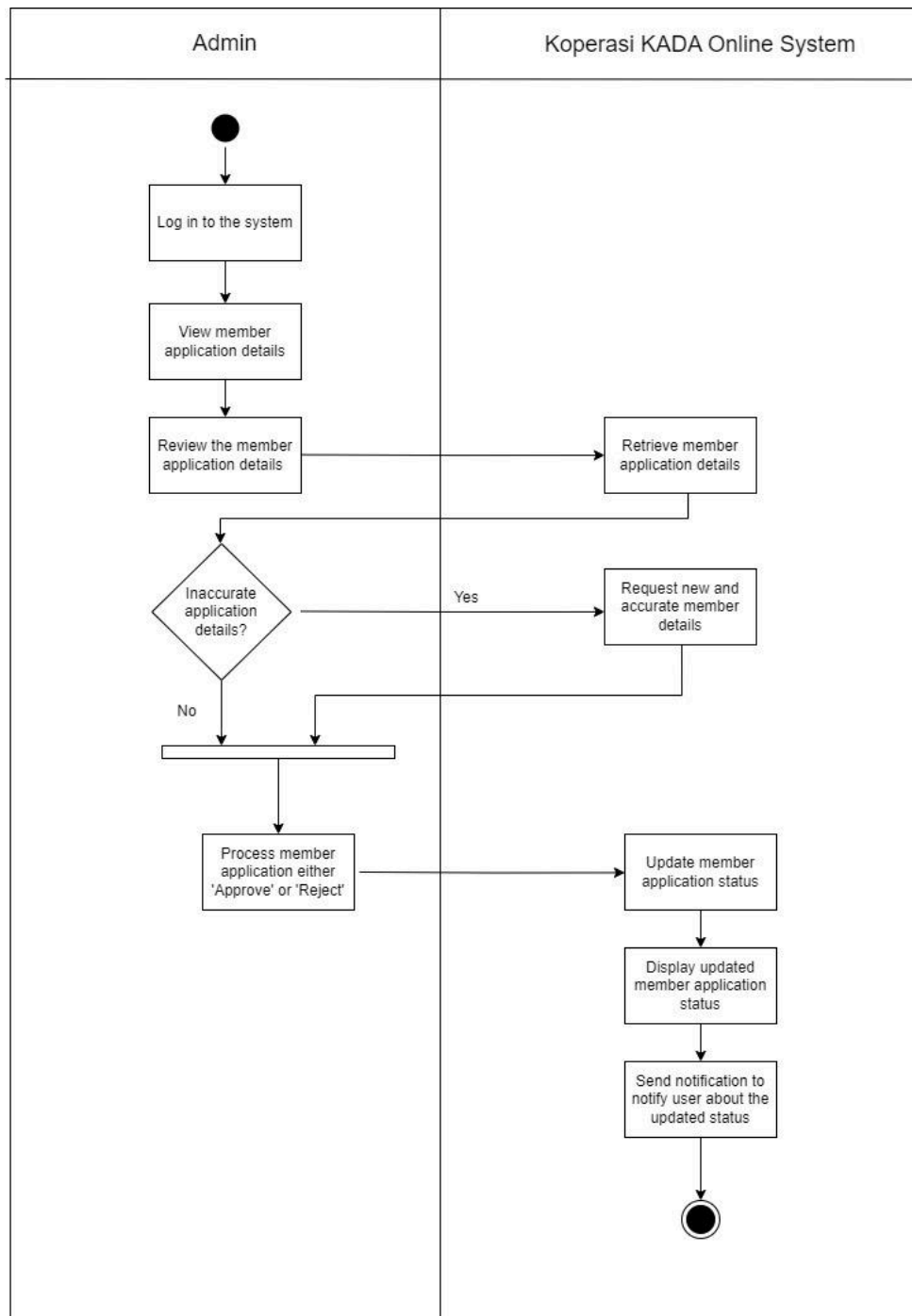


Figure 2.4.8.3: Activity Diagram for Approve Membership Application

2.4.9 UC009: Use Case < Approve Loan Application >

Table 2.4.9.1: User Story Description for Approve Loan Application

User story: Approve Loan Application
ID: US009
User Story Description As an Koperasi KADA admin I want to view and validate the members loan application information So that I can approve the Koperasi KADA members loan application.
Flow of events: 1. The admin logs into the system. 2. The admin selects the three status of the loan application progress, either approved, in progress and fail. 3. The system retrieves relevant data from the database, including new, in progress and completed loan applications. 4. The system generates an overall loan application report summary and detailed breakdown of each loan application information in a specified period. 5. The system displays the report, providing options to: view the report in detail, export the report (PDF, Excel) or print the report.
Alternative flow <i>n</i>: 1. If the applicant didn't provide the accurate information, the admin will ask the applicant to update new information.
Acceptance Criteria Precondition: 1. The admin logged in to the system. Postcondition: 1. Accurate and timely reports are generated. 2. Visual charts and summaries are easy to understand. 3. Exportable to formats like PDF or Excel.

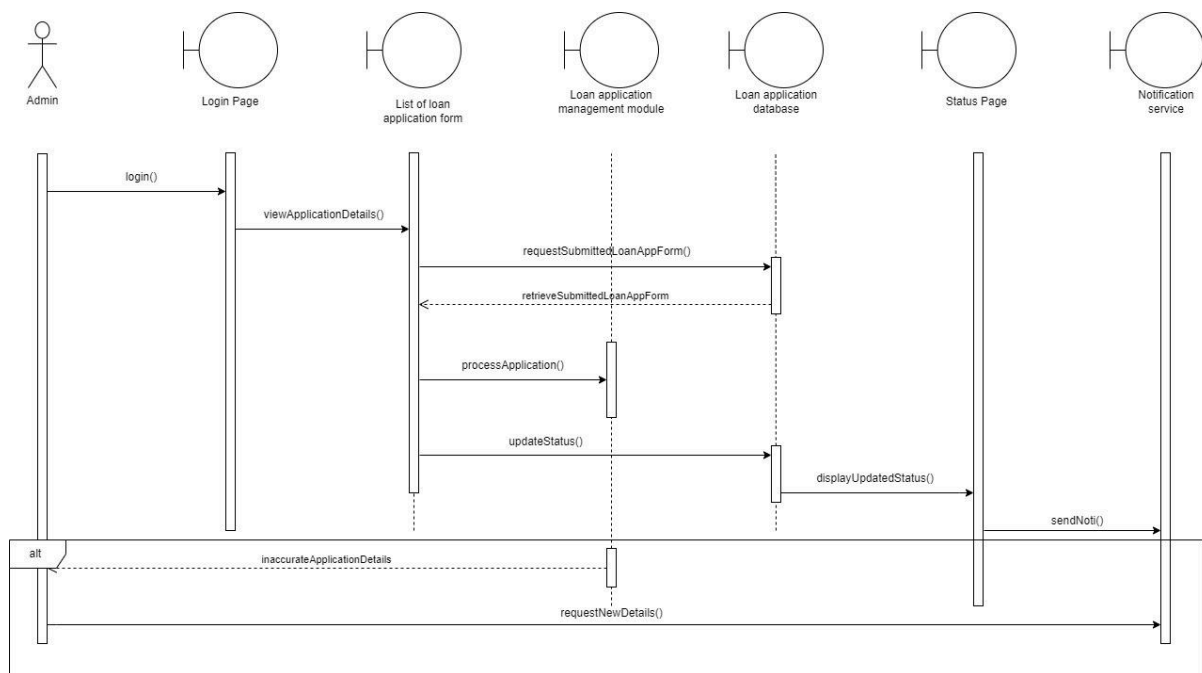


Figure 2.4.9.2: Sequence Diagram for Approve Loan Application

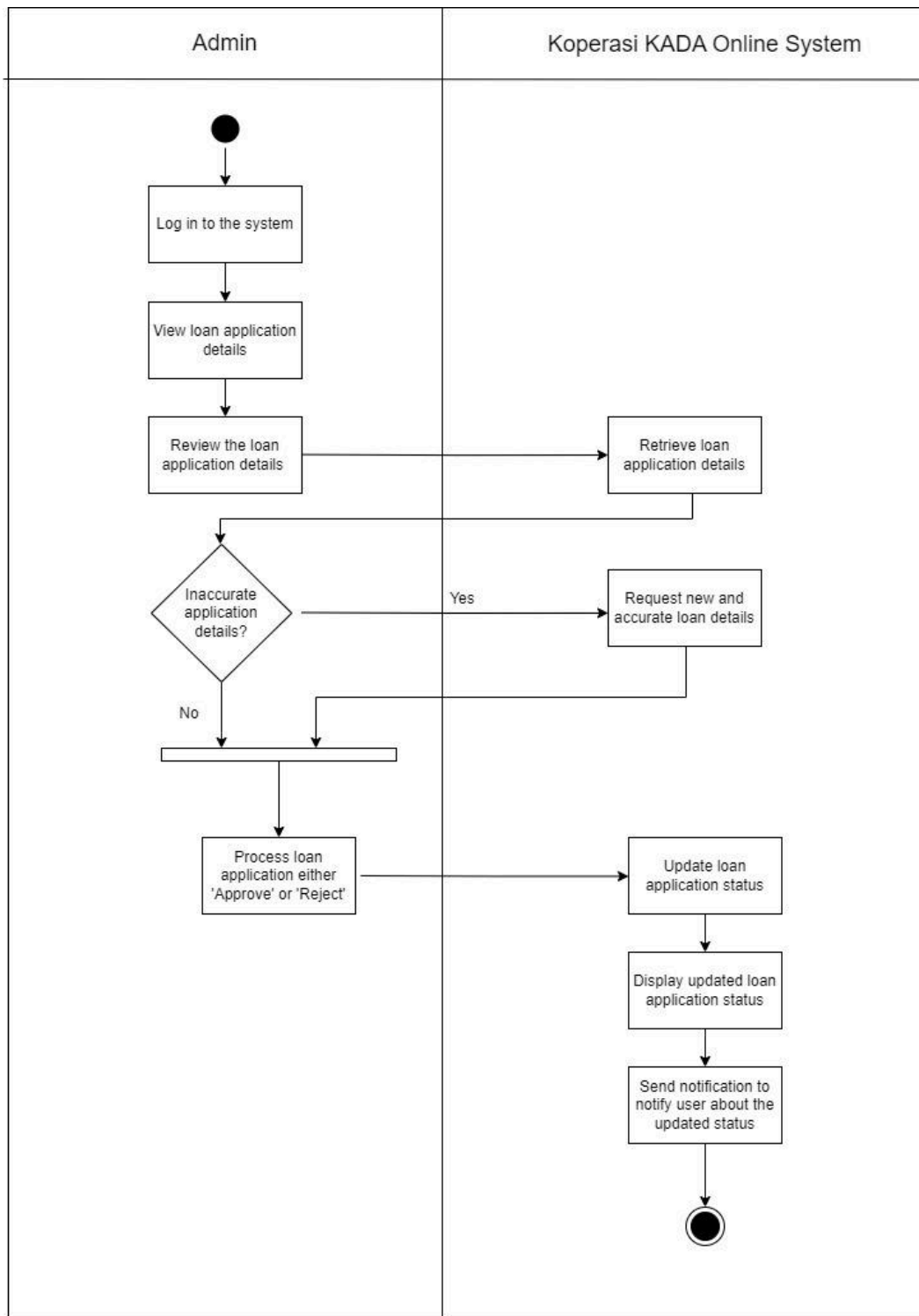


Figure 2.4.9.3: Activity Diagram for Approve Loan Application

2.4.10 UC0010: Use Case < Approve terminate membership application>

Table 2.4.10.1: User Story Description for Approve Terminate Membership Application

User story: Approve terminate membership application
ID: US010
User Story Description As an Koperasi KADA admin I want to review and validate the members' termination requests, So that I can approve or reject membership termination applications effectively.
Flow of events: 1. The admin logs into the system. 2. The admin selects the three status of the membership termination application progress, either approved, in progress and fail. 3. The system retrieves relevant data from the database, including new, in progress and completed membership termination applications. 4. The system generates an overall membership termination application report summary and detailed breakdown of each loan application information in a specified period. 5. The system displays the report, providing options to: view the report in detail, export the report (PDF, Excel) or print the report.
Alternative flow <i>n</i>: 1. If the applicant didn't provide the accurate information, the admin will ask the applicant to update new information.
Acceptance Criteria Precondition: 1. The admin logged in to the system. Postcondition: 1. Accurate and timely reports are generated. 2. Visual charts and summaries are easy to understand. 3. Exportable to formats like PDF or Excel.

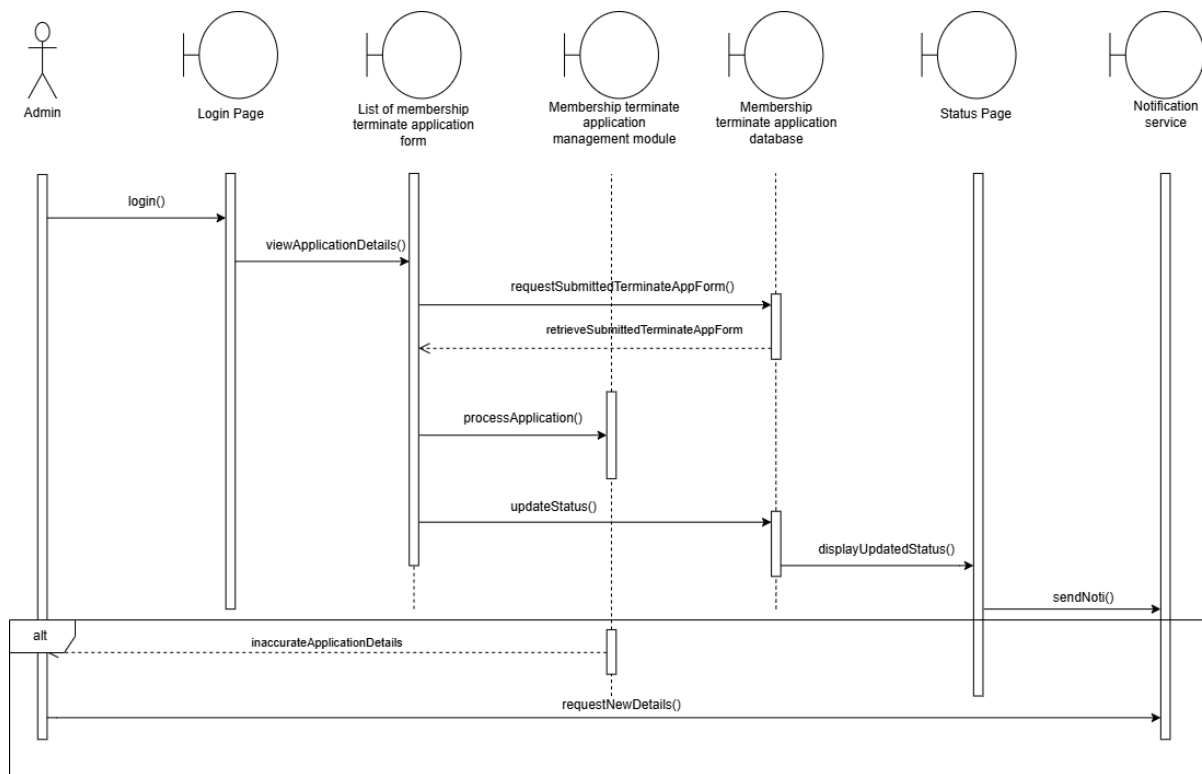


Figure 2.4.10.2: Sequence Diagram for Approve Terminate Membership Application

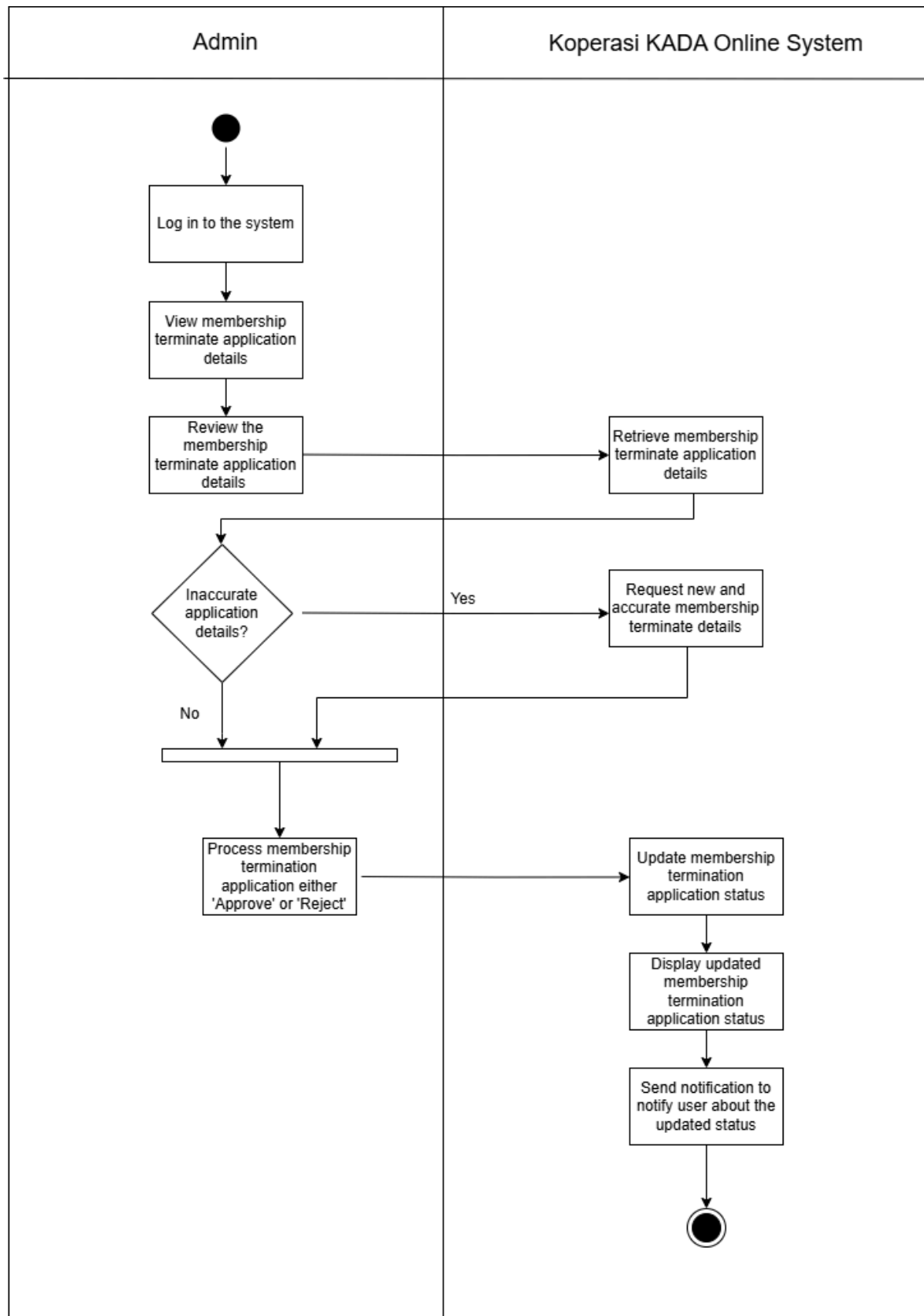


Figure 2.4.10.3: Activity Diagram for Approve Terminate Membership Application

2.4.11 UC0011: Use Case < Generate Report >

Table 2.4.11.1: User Story Description for Generate Report

User story: Generate Report
ID: US009
User Story Description As an Koperasi KADA admin I want to generate reports So that I can monitor the cooperative's membership and loan application growth.
Flow of events: 1. The admin logs into the system. 2. The admin selects the report type, either membership, loan or cooperation financial report. 3. The system retrieves relevant data from the database. 4. The system generates a financial summary and detailed breakdown of each category for the specified period. 5. The system displays the report, providing options to: view the report in detail, export the report (PDF, Excel) or print the report.
Alternative flow <i>n</i>: 1. If an invalid report type is detected, the system prompts the admin to select another correct type.
Acceptance Criteria Precondition: 1. The admin logged in to the system. Postcondition: 1. Accurate and timely reports are generated. 2. Visual charts and summaries are easy to understand. 3. Exportable to formats like PDF or Excel.

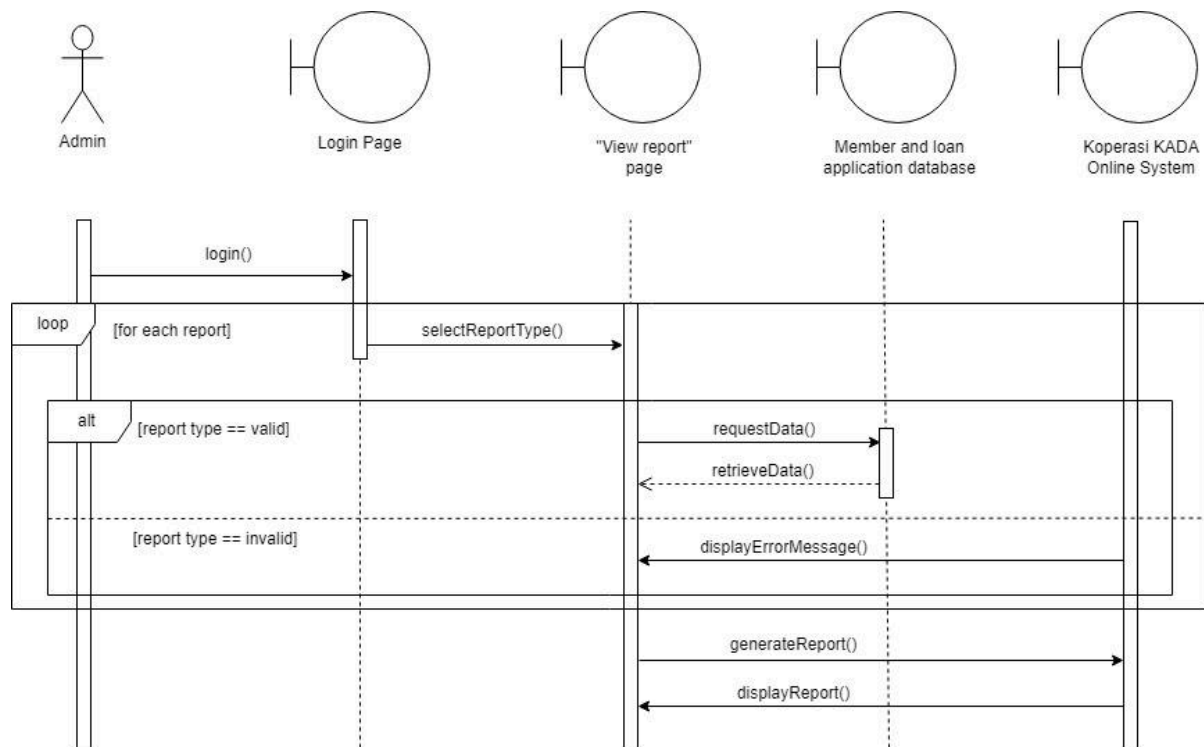


Figure 2.4.11.2: Sequence Diagram for Generate Report

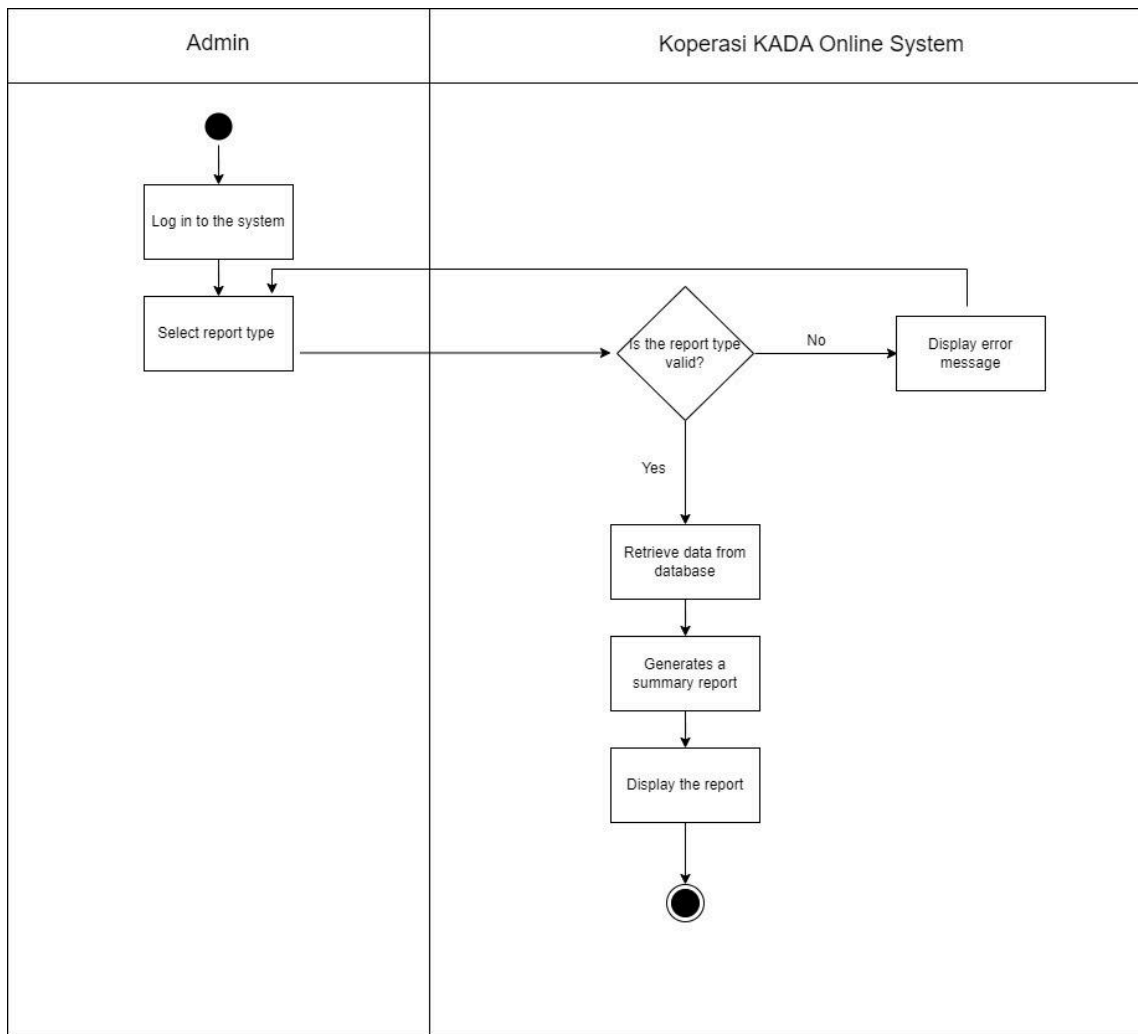


Figure 2.4.11.3: Activity Diagram for Generate Report

2.5 Performance and Other Requirements

In this section, we will identify the system's expected operational performance and how the system is supposed to behave.

Software system attributes define the overall qualities or characteristics of the system. These attributes are the foundation of the Koperasi KADA Online System, as listed below:

- Usability: The system should be simple for the user to navigate and interact with the UI and other functionalities.
- Reliability: The system should be able to perform its functions correctly and consistently, while also being able to handle any errors or system failures.
- Maintainability: The system can be modified, repaired, and enhanced to improve its scalability.
- Compatibility: The system is able to work with most OS systems such as Windows, macOS, Linux and mobile devices like Android and iOS. The system should be able to adapt to different interfaces, such as desktop interface and mobile interface,

Performance requirements define how well the system must perform under certain conditions to meet user expectations. The requirements are listed below:

- Response Time: The system should be able to minimize response time to under 1 second.
- Real-Time Synchronization: The system should be able to actively update user data in real-time without any interruptions.
- Capacity: The system should be able to support up to 100 people at a time. This is to maintain the systems' operationality during peak usage.
- Availability: The system should be available for use 99.9% of the operational time. Any system failure should be immediately addressed to reduce the delay of system operational time.

Other requirements refer to any non-functional requirements that define the system's qualities, separate from the system's functionality. These requirements include the following:

- Security: The system should be able to provide secure data access for the admin. It should also prevent potential data breaches from unauthorized access.
- Safety: The system should be able to operate safely without causing harm or crashes to the user's environment.
- Legal and Regulatory: The system's policy should comply with standard laws, regulations and standards.
- Localisation: The system should follow local standards (Malaysia) such as time zone (UTC+8) and money currency (RM).

2.6 Design Constraints

Security Compliance: The system must comply with data protection regulations such as GDPR (General Data Protection Regulation) and PDPA. To protect member information, Two-Factor Authentication (2FA) must be used for secure user access.

Data Integrity: All sensitive data must be accurately processed and stored, with redundancy in place to reduce the risk of data loss. Periodic data integrity checks should be performed to prevent unauthorised changes or corruption.

Scalability: The system's infrastructure should be scalable to accommodate an expanding user base. This includes a cloud-based solution that can handle increasing workloads while maintaining high availability during peak usage periods.

Cross-Browser Compatibility: The system must work seamlessly with popular web browsers like Google Chrome, Mozilla Firefox, and Microsoft Edge, as well as on a variety of devices like desktops, tablets, and smartphones.

Role-Based Access Control: The system must enforce Role-Based Access Control (RBAC), which ensures that permissions and access levels are tailored to specific user roles (for example, administrators, clients, members, and board directors). This includes limiting sensitive functionalities to only authorised personnel.